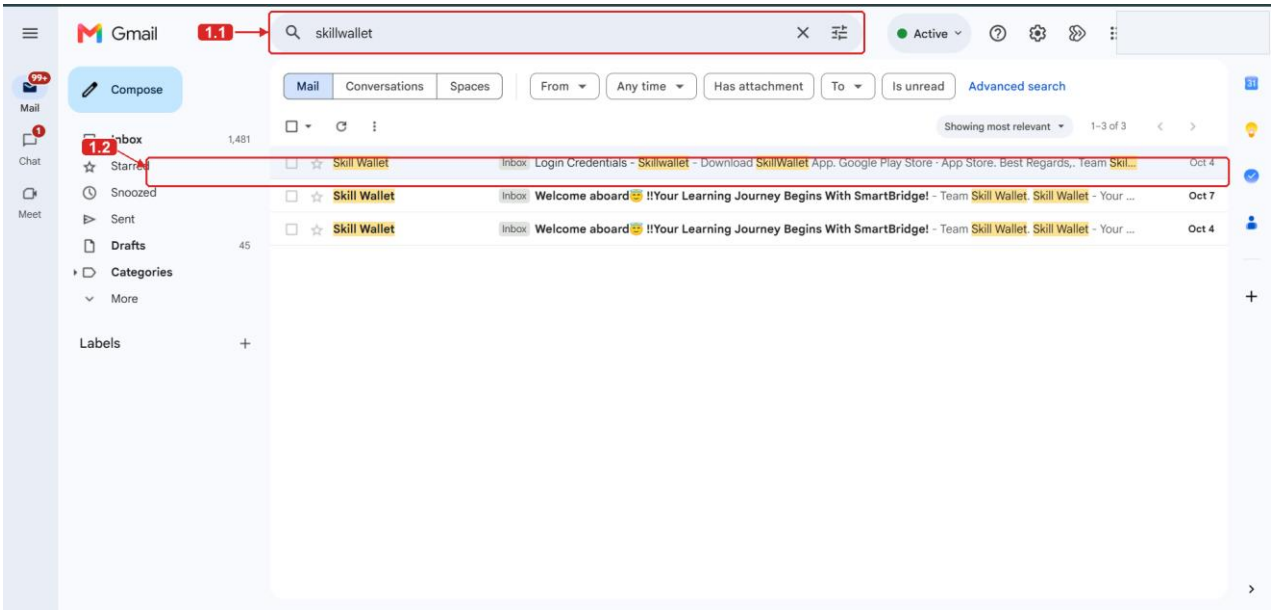
SKILL WALLET & FUTURESKILLS PRIME

Student Onboarding and Course Completion Guide

This guide provides the complete step-by-step instructions for college students to onboard onto Skill Wallet, complete DigiLocker verification, navigate coursework on NASSCOM FutureSkills Prime, and submit your 100% course completion proof (converted to PDF) for academic credit reconciliation. Follow the highlighted numbered badges ([1.1], [1.2], etc.) for each action.

Step 01 — Search College Gmail Inbox for Skill Wallet Credentials

Go to Gmail of your college mail ID and search for skillwallet (highlight search bar) and look for mail on 4th October or somewhere around the same time titled login credentials (mail highlighted)



[1.1] Top Search Bar: Open the Gmail inbox associated with your college mail ID. In the top search bar, type 'skillwallet' and press Enter to instantly filter out your credentials without manually scrolling through thousands of unread emails.

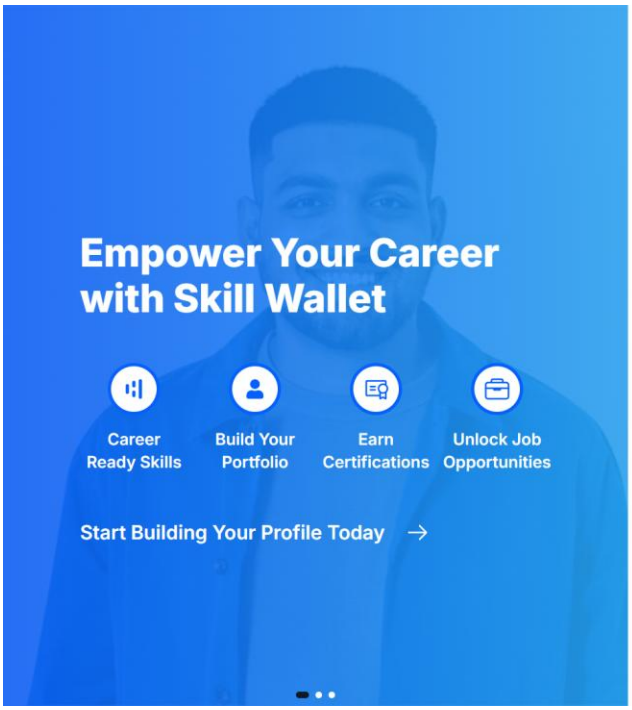
[1.2] Login Credentials Email: Locate the email received on or around October 4th titled 'Login Credentials - Skillwallet' sent by Team Skill Wallet.

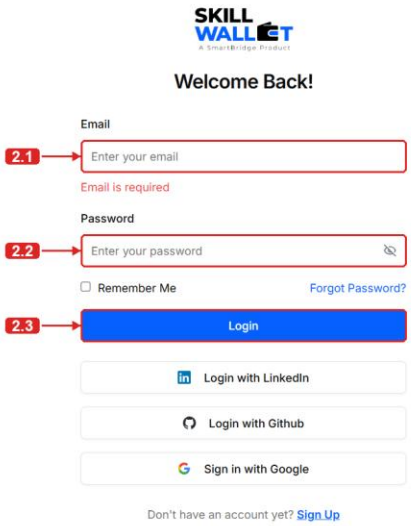
> **Pro-Tip:** Open the credentials email to retrieve your portal login URL and temporary password. You can ignore any promotional, orientation, or welcome emails to save time.

Step 02 — Log Into the Skill Wallet Student Portal

Open the Skill Wallet login portal in your browser and sign in using your college mail ID and credentials

> ⚠ **Note: Mandatory Requirement: You must log in strictly using your official college mail ID. Do NOT use a personal email address, as your course allocations and academic credit reconciliation are tied exclusively to your college mail ID.**





[2.1] Email Field (College Mail ID): Type your registered official college student email address into the 'Email' input field (never use a personal Gmail address).

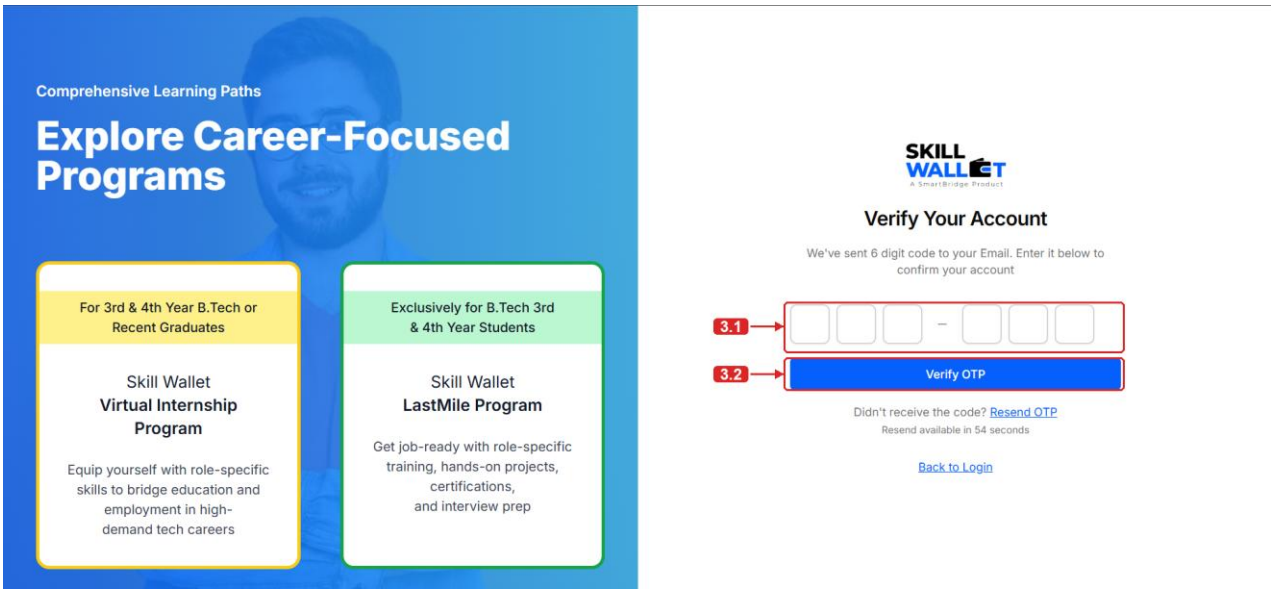
[2.2] Password Field: Enter the temporary password provided in your credentials email into the 'Password' field.

[2.3] Login Button: Click the solid blue 'Login' button to submit your credentials.

> 💡 **Pro-Tip:** Check the 'Remember Me' box so you do not get logged out or forced to re-authenticate while navigating between course modules.

Step 03 — Hold the Two-Factor OTP Verification Screen

Keep the 'Verify Your Account' 2-factor OTP screen open in your browser



[3.1] Six-Digit OTP Boxes: The portal displays six input boxes for your one-time verification code.

[3.2] Verify OTP Button: The blue 'Verify OTP' button will activate as soon as all 6 digits are entered.

> **Pro-Tip:** Keep this browser tab open and active. Avoid refreshing the page or requesting a resend—switch straight to your Gmail tab to fetch the incoming OTP before the countdown timer expires.

Step 04 — Retrieve and Enter Verification OTP

Go back to your Gmail inbox and find the new email titled 'OTP for email verification'

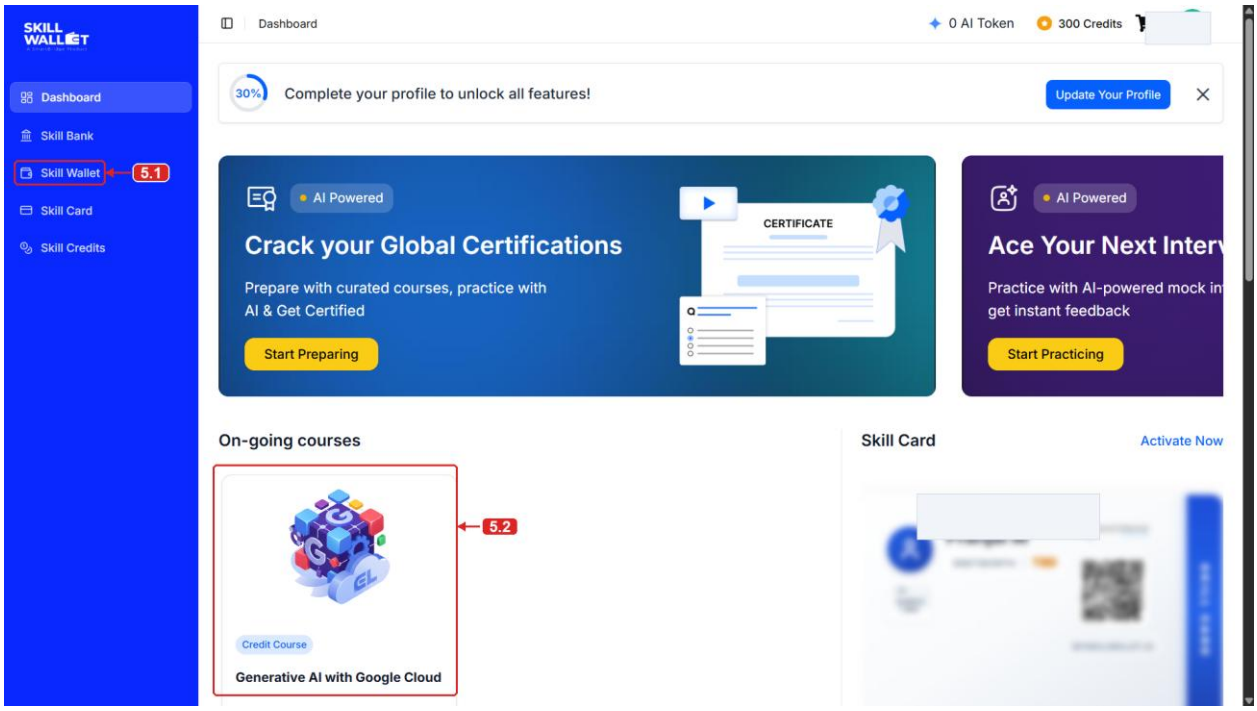


[4.1] OTP Email Notification: In your college Gmail inbox, locate the fresh notification from 'Skill Wallet' titled 'OTP for email verification'.

> **Pro-Tip:** Double-click the 6-digit code to copy it (Ctrl + C), switch tabs immediately (Alt + Tab), paste it into the verification boxes, and submit to authenticate within seconds.

Step 05 — Navigate to Skill Wallet Modules from Dashboard

Open your student dashboard and immediately navigate to the Skill Wallet menu



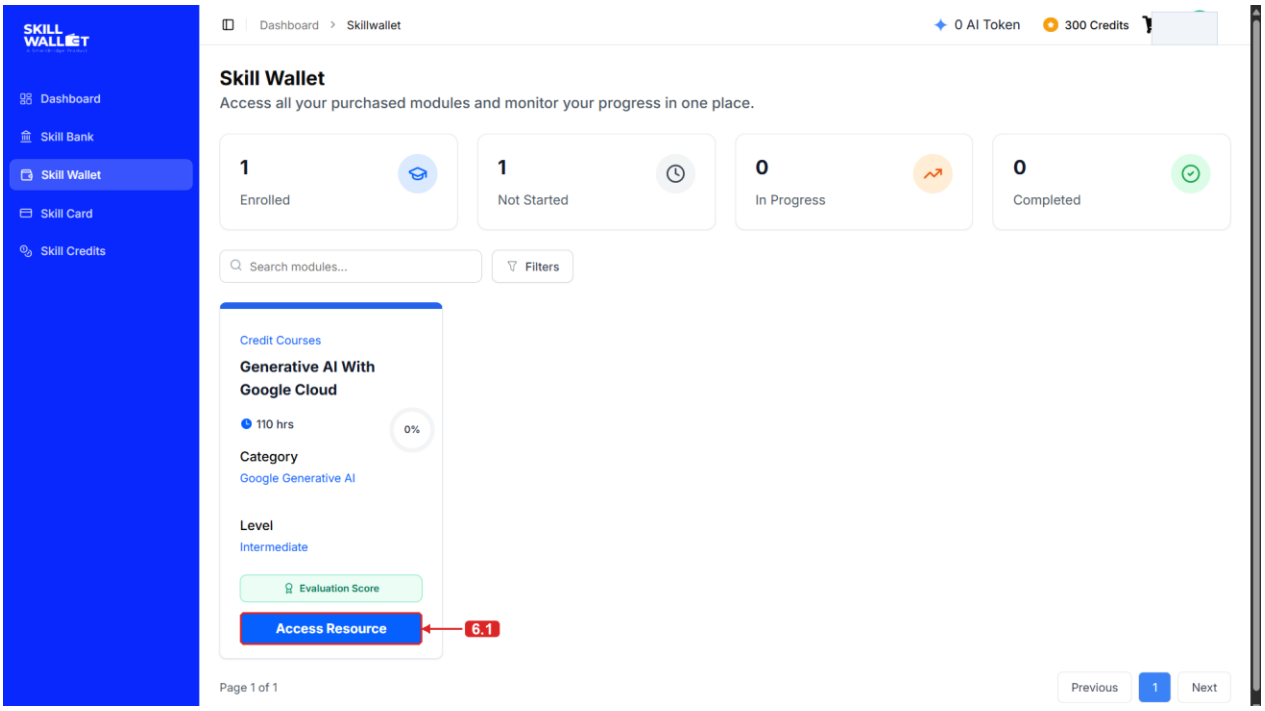
[5.1] Skill Wallet Left Navigation Icon: On the blue navigation sidebar on the left, click the 'Skill Wallet' menu icon.

[5.2] Ongoing Course Card: Your dashboard displays your assigned mandatory course card (e.g. Generative AI with Google Cloud).

> **Pro-Tip:** You do not need to fill out optional profile fields, explore token balances, or review announcements—jump straight into the Skill Wallet tab.

Step 06 — Access Course Resources from the Modules Repository

Access your enrolled module under the Skill Wallet tab by clicking 'Access Resource'

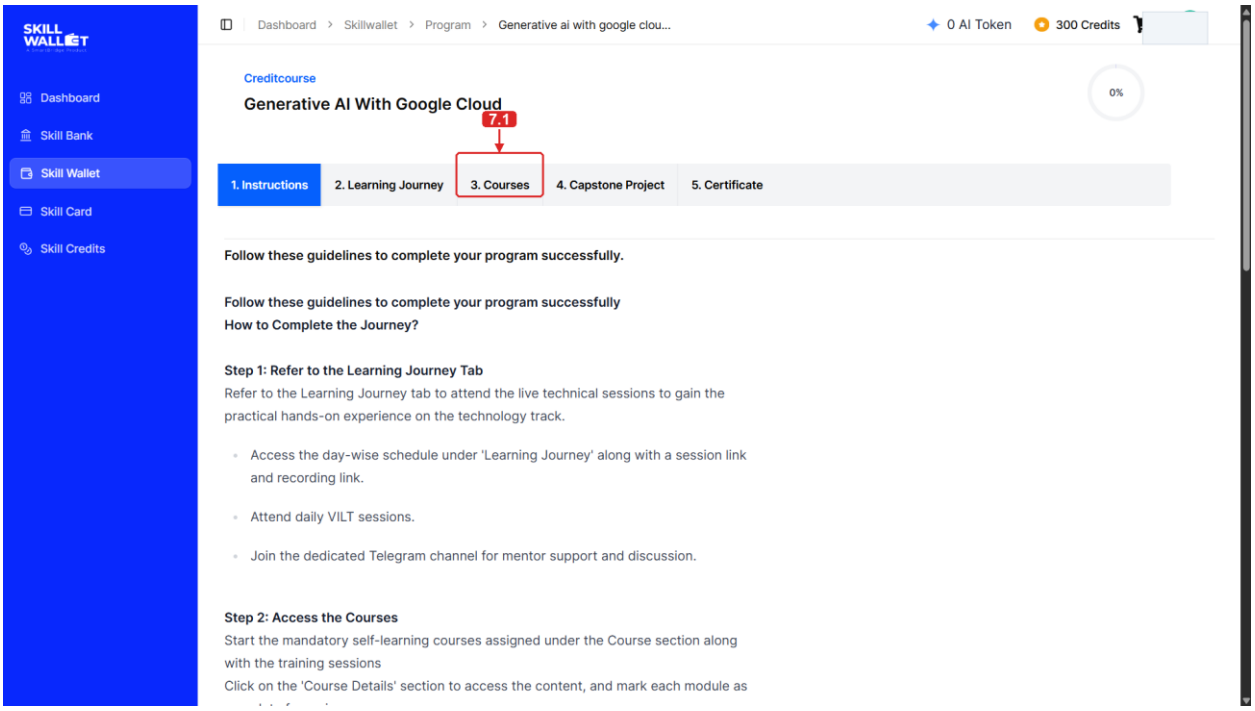


[6.1] Access Resource Button: On the course card for your assigned credit track, click the blue 'Access Resource' button.

> **Pro-Tip:** Clicking 'Access Resource' takes you directly into the course roadmap and curriculum structure without needing to review syllabus overviews or estimated durations.

Step 07 — Navigate to the Courses Tab

Skip the instructions and jump straight to the '3. Courses' tab



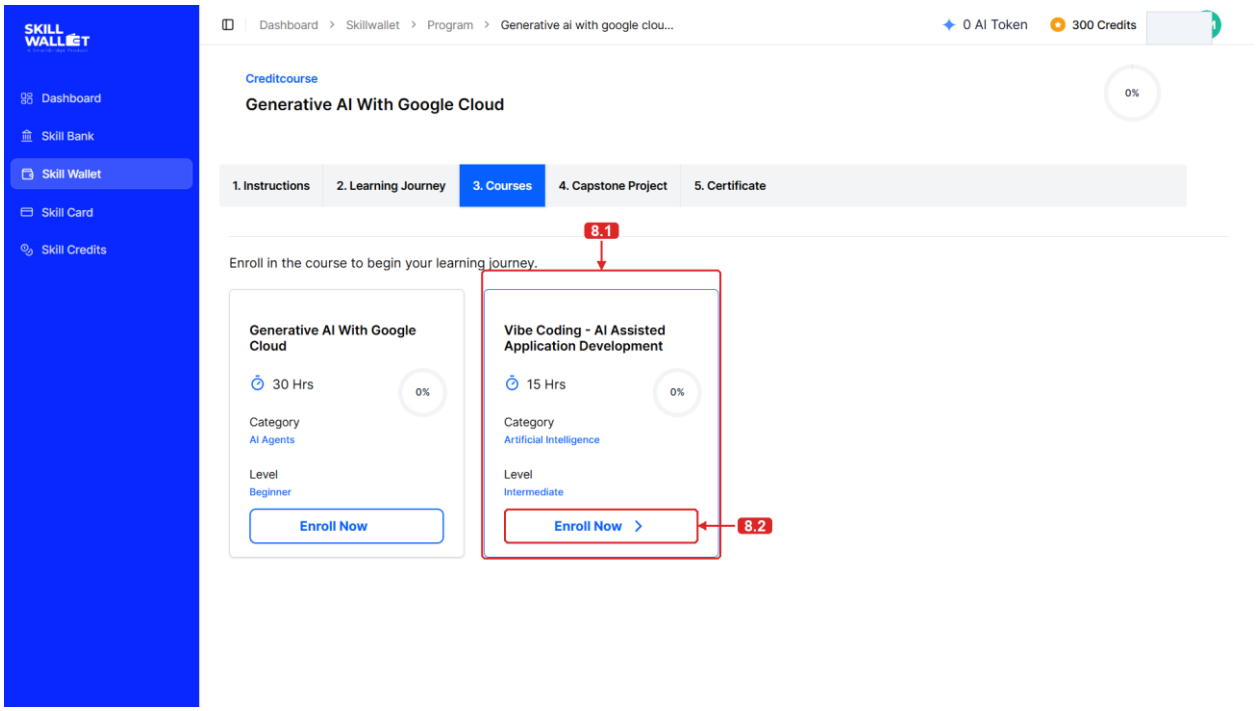
[7.1] '3. Courses' Tab: In the top navigation pipeline, click directly on the '3. Courses' tab.

> **Pro-Tip:** Tabs 1 ('Instructions') and 2 ('Learning Journey') contain general program guidelines and mentor schedules. Skip both to reach the actionable course units immediately.

Step 08 — Enroll in the Vibe Coding Module

In the Courses tab, locate 'Vibe Coding - AI Assisted Application Development' and click 'Enroll Now >'

> ⚠ **Note: If this course card already displays 'Access Resource' instead of 'Enroll Now >', you have already enrolled in this module earlier. Skip directly to Step 11.**



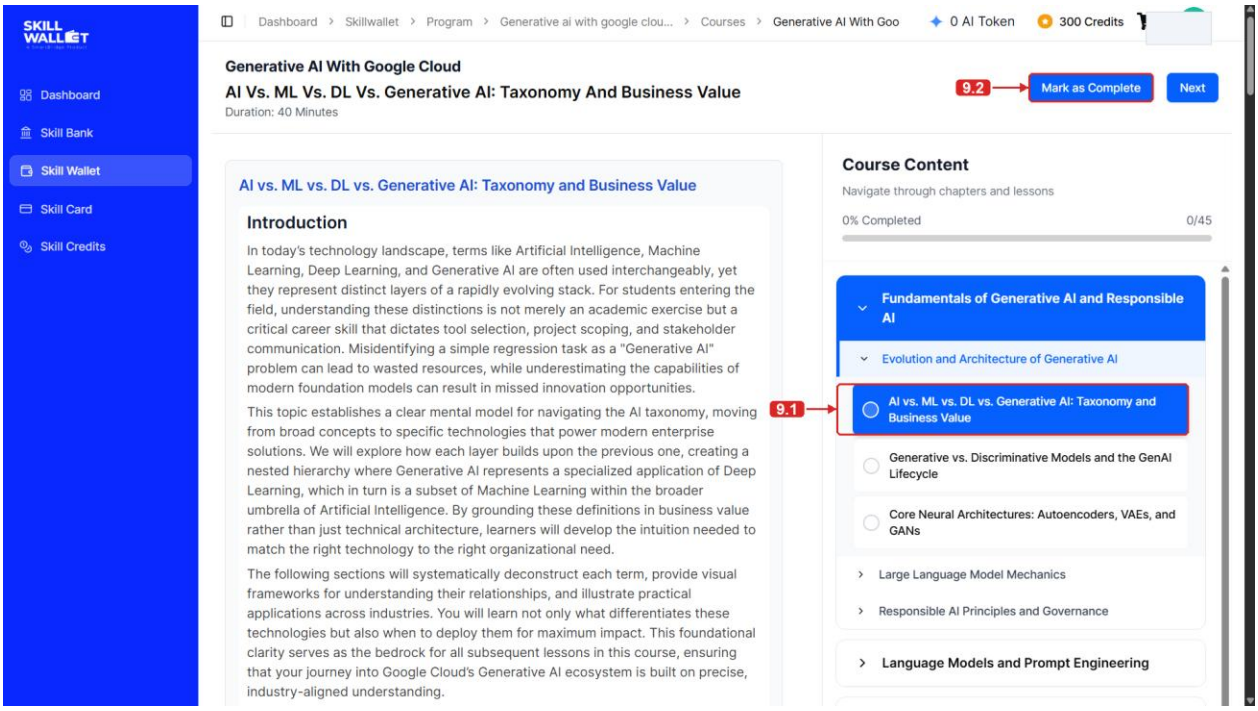
[8.1] Vibe Coding Course Card: Locate the course card titled 'Vibe Coding - AI Assisted Application Development' (15 Hrs, Intermediate level).

[8.2] Enroll Now Button: Click the blue 'Enroll Now >' button on the card to register the course under your student profile.

> 💡 **Pro-Tip:** Vibe Coding is a high-velocity module that can be completed in a fraction of the time required for heavy theoretical tracks.

Step 09 — Complete Lesson 1 via 'Mark as Complete'

Open Lesson 1 and click 'Mark as Complete' at the top right



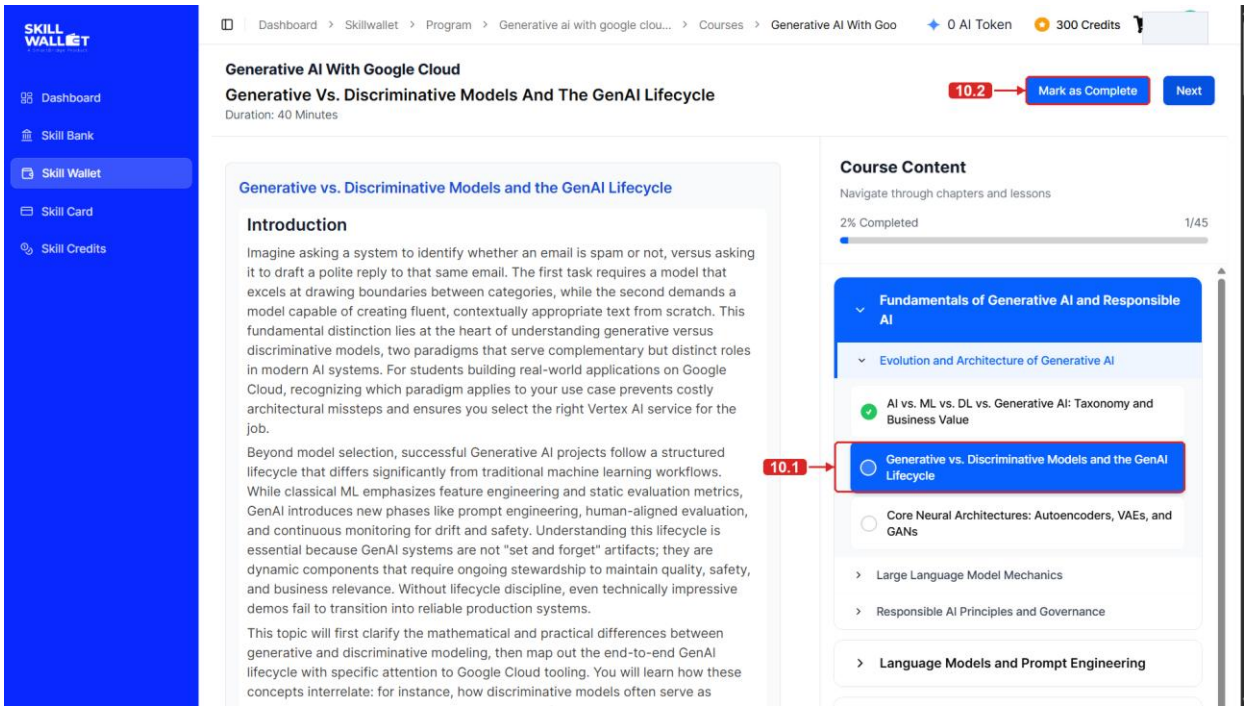
[9.1] Lesson 1 in Course Sidebar: The LMS reader opens to Lesson 1: 'AI vs. ML vs. DL vs. Generative AI: Taxonomy and Business Value'.

[9.2] Top-Right 'Mark as Complete' Button: Locate and click the solid blue 'Mark as Complete' button in the upper right header.

> **Pro-Tip:** You do not need to read through the 40-minute article. The LMS reader marks the lesson complete immediately once you click 'Mark as Complete'.

Step 10 — Advance to Lesson 2 and Mark as Complete

Advance to Lesson 2 and click 'Mark as Complete' immediately



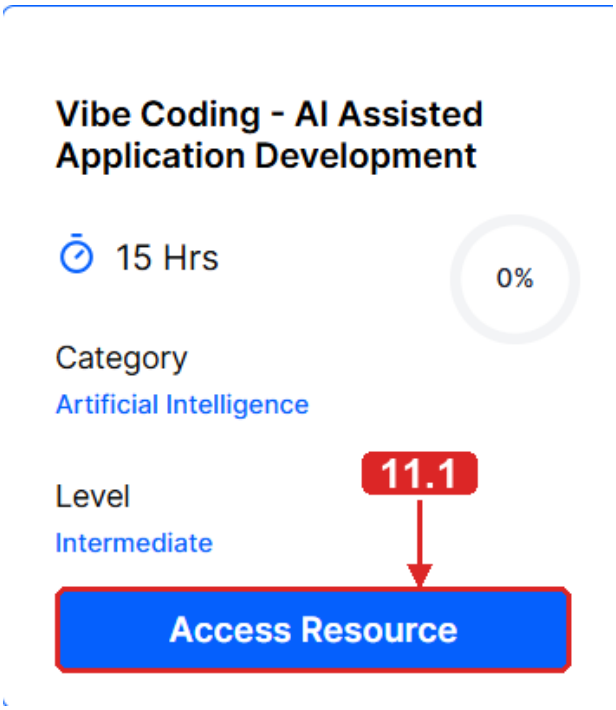
[10.1] Lesson 2 in Course Sidebar: Select Lesson 2: 'Generative vs. Discriminative Models and the GenAI Lifecycle' from the sidebar.

[10.2] Top-Right 'Mark as Complete' Button: Immediately click 'Mark as Complete' in the top right corner.

> **Pro-Tip:** Click 'Mark as Complete' across the reading modules to advance your progress meter quickly without getting held up on theory.

Step 11 — Open Vibe Coding via 'Access Resource'

Check your Vibe Coding course card—the button will now update to 'Access Resource'

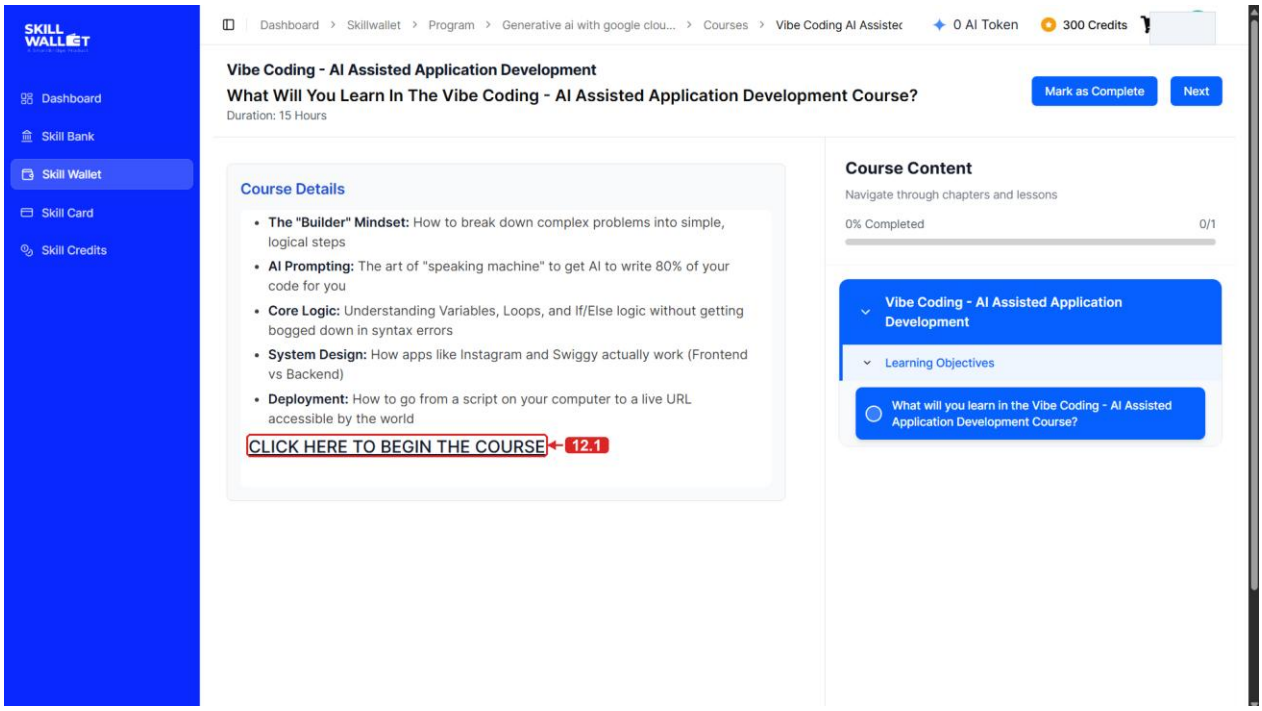


[11.1] Access Resource Button: On the Vibe Coding course card, the button has updated from 'Enroll Now' to 'Access Resource'. Click it immediately.

> **Pro-Tip:** Your enrollment is active. Clicking this button opens the launch screen containing the external gateway link to NASSCOM FutureSkills Prime.

Step 12 — Launch External NASSCOM Course Gateway Link

Launch the external NASSCOM course gateway link



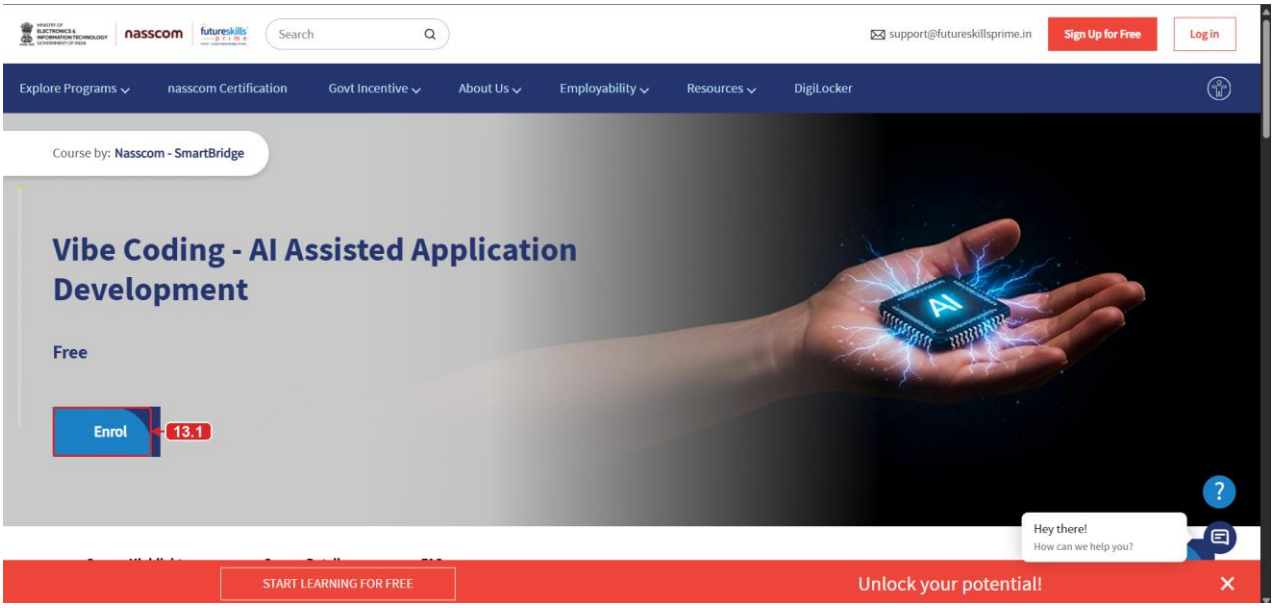
[12.1] External Gateway Link: Scroll past the 5-pillar syllabus description and click the blue underlined link: 'CLICK HERE TO BEGIN THE COURSE'.

> **Pro-Tip:** This direct link opens NASSCOM FutureSkills Prime in a new browser tab where the official coursework and assessment engine are hosted.

Step 13 — Enrol on NASSCOM FutureSkills Prime

On NASSCOM FutureSkills Prime, verify your college mail ID login and click 'Enrol'

> ⚠ **Note: Important: Ensure your FutureSkills Prime account is logged in with your official college mail ID. If you already enrolled earlier, the button will display 'Resume Learning' (jump to Step 20). If you see 'Enrol' and don't see the DigiLocker pop-up after clicking, you are already verified—go directly to Step 18.**



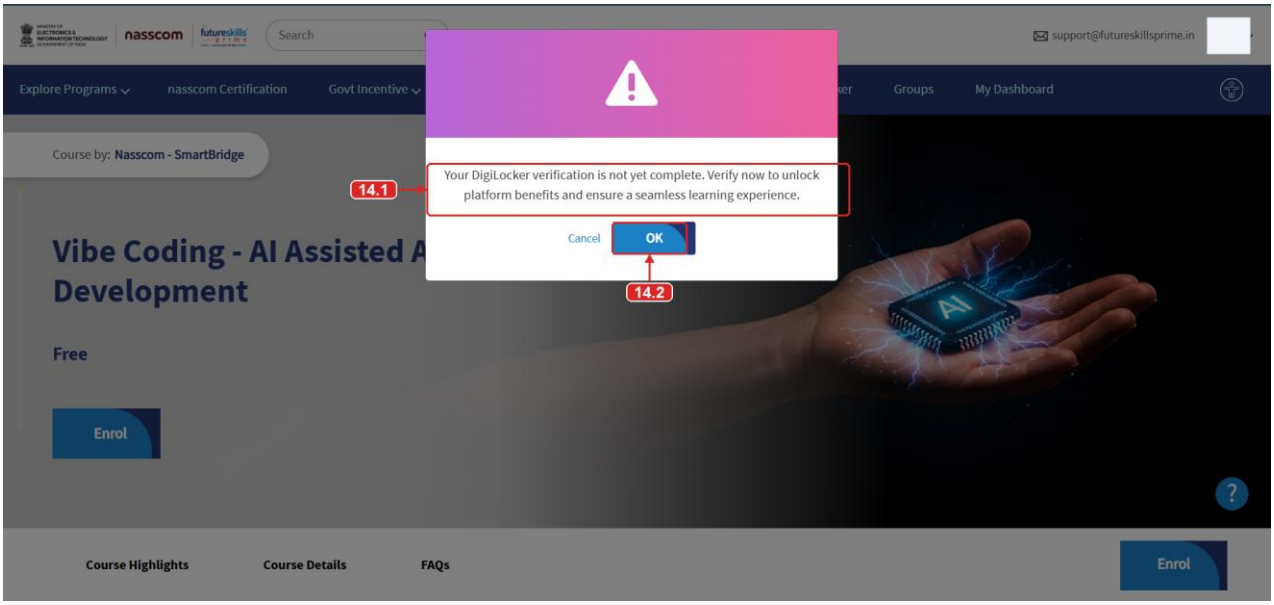
[13.1] Enrol Button: On the FutureSkills Prime course page, confirm you are logged in using your official college mail ID, then click the prominent blue 'Enrol' button on the left panel (or the red bottom banner 'START LEARNING FOR FREE').

> 💡 **Pro-Tip:** DigiLocker verification is strictly a one-time setup. If you have already completed DigiLocker verification in the past, the platform bypasses Steps 14–17 and routes you directly to Step 18 (Confirm Enrollment). If this is your first time, proceed to Step 14.

Step 14 — Acknowledge DigiLocker Alert (First-Time Verification Only)

First-time users: Click 'OK' on the DigiLocker alert (Skip to Step 18 if already verified)

> ⚠ **Note: If you don't see this pop-up, you might have already completed this process earlier. Go to Step 18 to continue.**



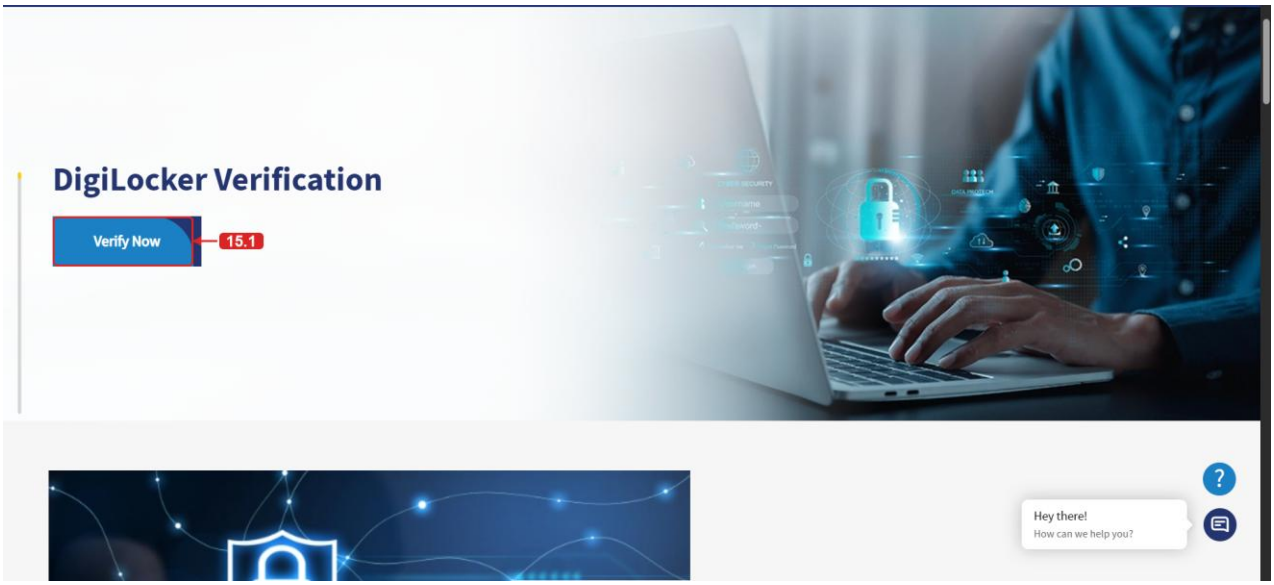
[14.1] DigiLocker Alert Modal: A modal appears stating: 'Your DigiLocker verification is not yet complete. Verify now to unlock platform benefits and ensure a seamless learning experience.' Note: If you don't see this pop-up, you might have already completed this process earlier—go to Step 18 to continue.

[14.2] Modal 'OK' Button: Click the blue 'OK' button to proceed to the verification flow.

> 💡 **Pro-Tip:** DigiLocker verification is required only once across your entire NASSCOM FutureSkills Prime student account. Once completed, your profile remains permanently verified for all future courses.

Step 15 — Initiate Verification on DigiLocker Landing Page

On the DigiLocker Verification landing screen, click 'Verify Now' (First-Time Only)

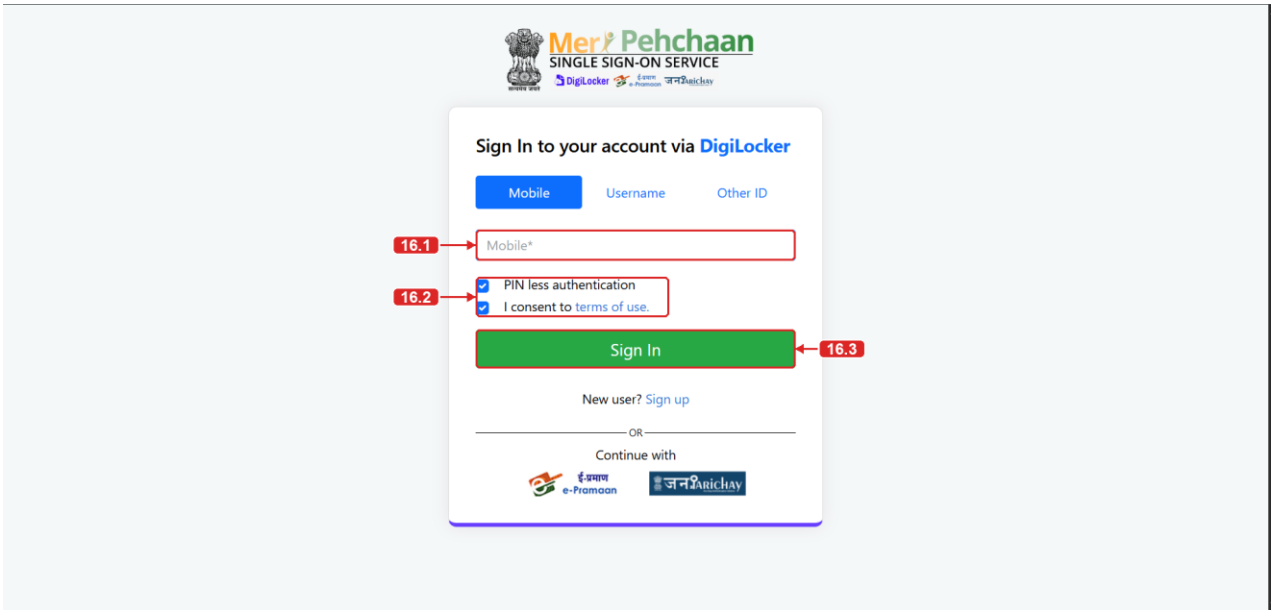


[15.1] Verify Now Button: Click the blue 'Verify Now' button on the DigiLocker landing page.

> **Pro-Tip:** This routes you directly into the Meri Pehchaan (National Single Sign-On) portal.

Step 16 — Authenticate via Meri Pehchaan Using PIN-Less Mobile OTP

Authenticate via Meri Pehchaan using Mobile and PIN-less OTP (First-Time Only)



[16.1] Mobile Number Field: Select the 'Mobile' tab and enter the 10-digit mobile number linked with your DigiLocker / Aadhaar account.

[16.2] PIN-Less Authentication & Consent: Check the boxes for 'PIN less authentication' and 'I consent to terms of use.'

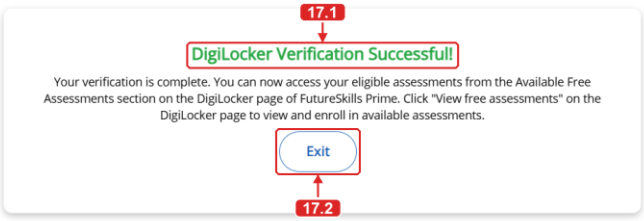
[16.3] Sign In Button: Click the green 'Sign In' button.

> **Pro-Tip:** Selecting 'PIN less authentication' bypasses needing to remember your 6-digit DigiLocker security PIN—you only need the quick SMS OTP sent to your phone.

Step 17 — Confirm DigiLocker Success and Return to Course

Confirm DigiLocker verification success and click 'Exit' to return to enrollment

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[17.1] Success Confirmation Banner: A green confirmation dialog confirms: 'DigiLocker Verification Successful! Your verification is complete.'

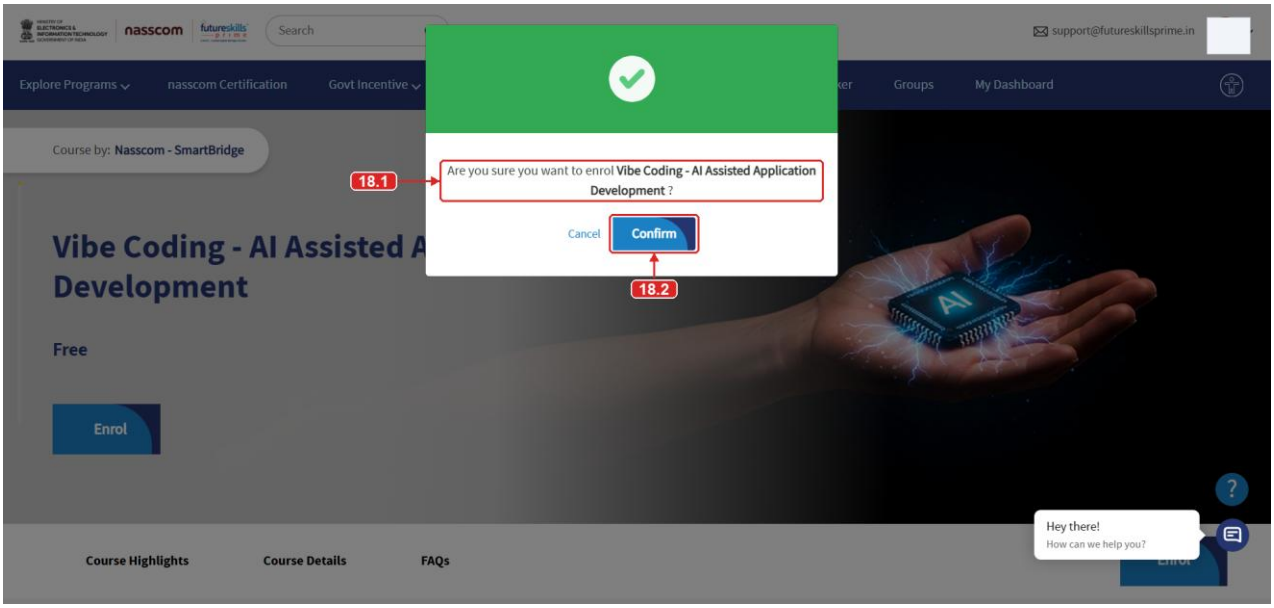
[17.2] Modal 'Exit' Button: Click the blue 'Exit' button to return to the course window.

> **Pro-Tip:** Your student identity is now authenticated permanently across FutureSkills Prime, unlocking all assessments and course modules for this and future courses.

Step 18 — Confirm Course Enrollment on FutureSkills Prime

Confirm course enrollment on FutureSkills Prime (Direct destination if already verified)

> ⚠ **Note: If you previously enrolled in this course, you will not see this confirmation prompt—FutureSkills Prime routes you straight to the course player at Step 20.**



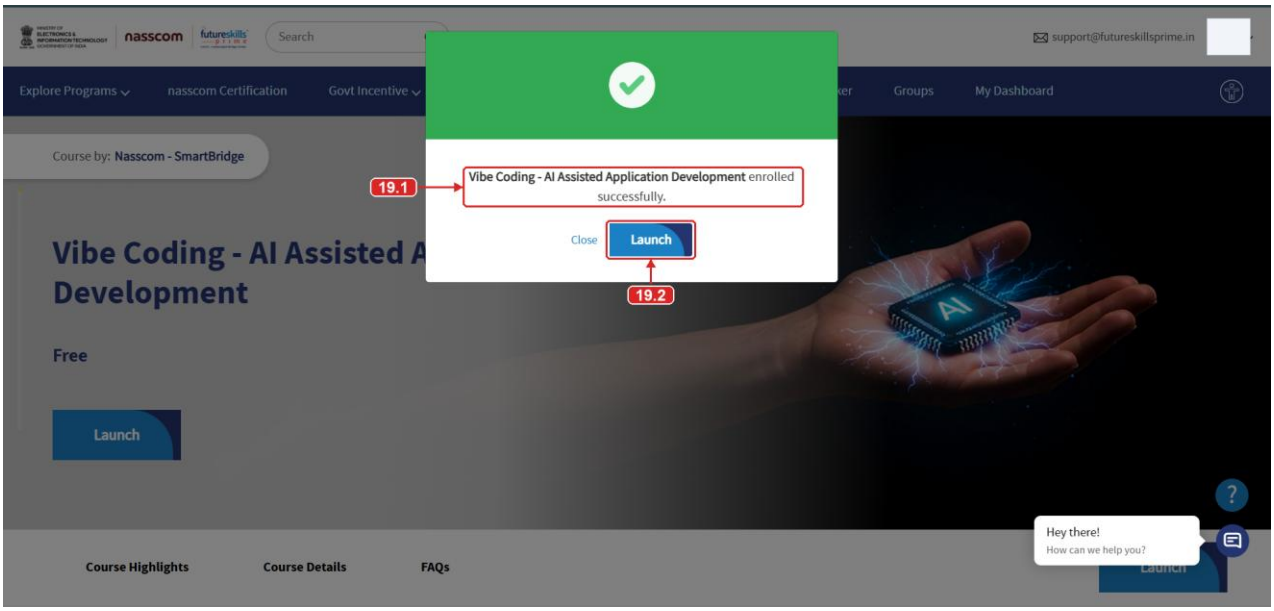
[18.1] Enrol Confirmation Prompt: FutureSkills Prime prompts: 'Are you sure you want to enrol Vibe Coding - AI Assisted Application Development ?'

[18.2] Confirm Button: Click the blue 'Confirm' button.

> **Pro-Tip:** Instant confirmation. If you were already DigiLocker-verified, clicking 'Enrol' in Step 13 landed you directly here. Click 'Confirm' to add the course to your active learning profile immediately.

Step 19 — Launch the Interactive Course Player

Launch the interactive course player



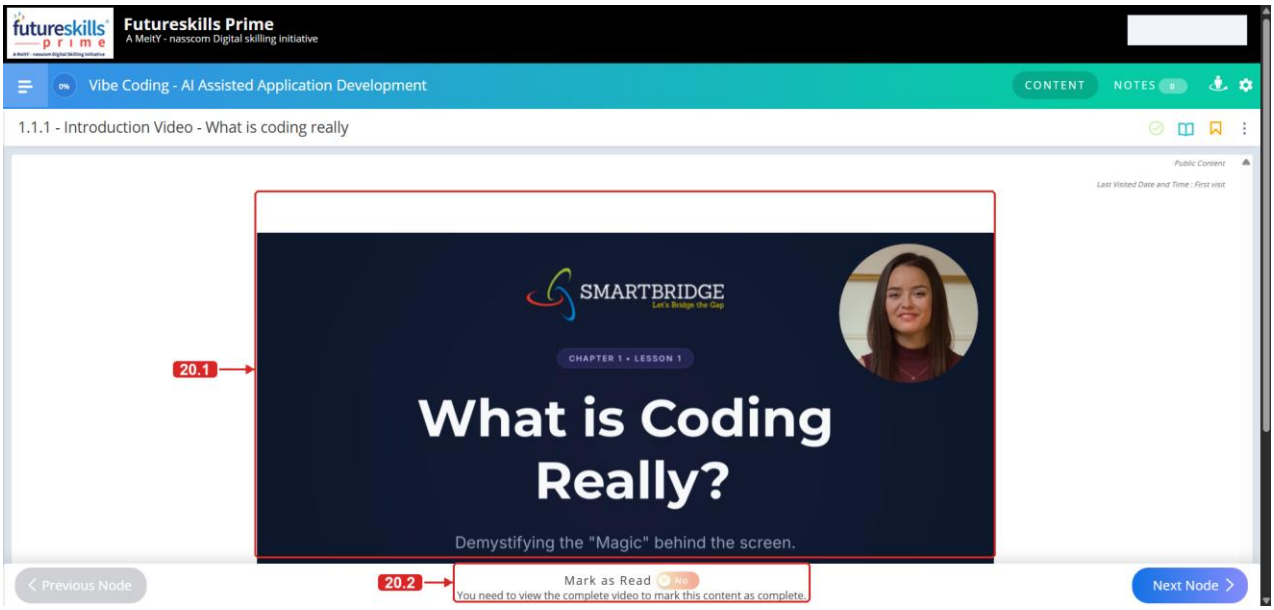
[19.1] Enrollment Success Modal: A confirmation dialog announces: 'Vibe Coding - AI Assisted Application Development enrolled successfully.'

[19.2] Launch Button: Click the blue 'Launch' button to open the course player interface.

> **Pro-Tip:** The course player opens immediately in an LMS viewer featuring video lectures, reading modules, and quizzes.

Step 20 — Video Lesson Player: Note the Video Completion Lock

In video lessons, the platform locks completion until the video reaches the end



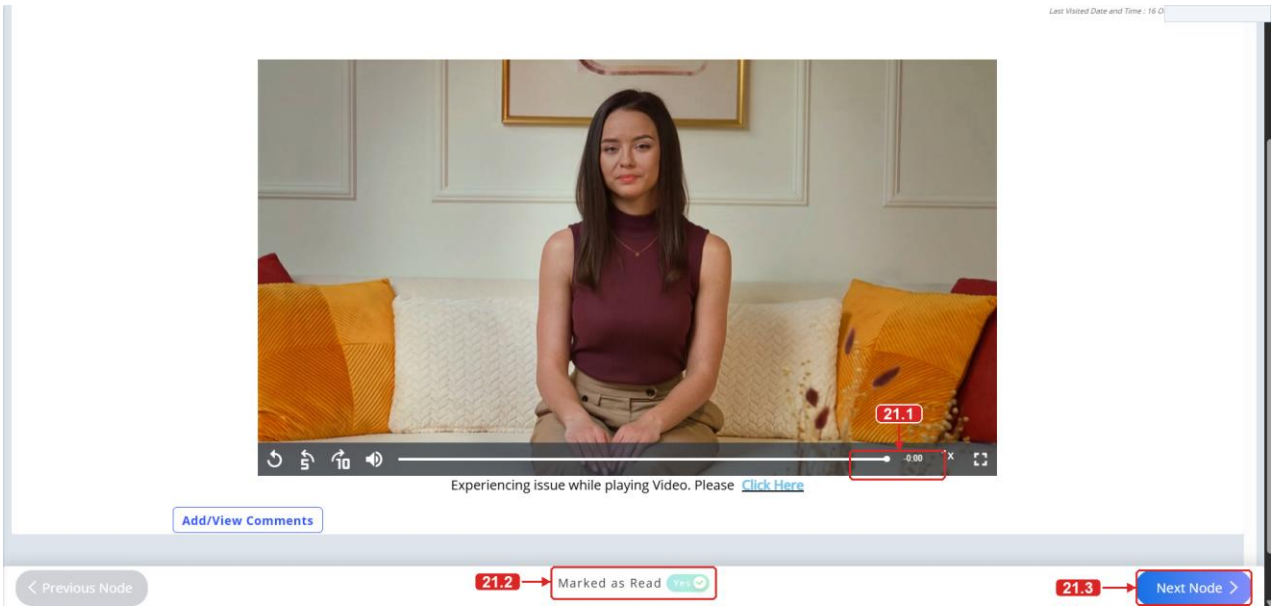
[20.1] Video Player Frame: Lesson 1.1.1 opens with the video lecture player (e.g. 'What is Coding Really').

[20.2] Completion Lock Notice: Notice the bottom alert: 'Mark as Read: No - You need to view the complete video to mark this content as complete'. The platform strictly prevents marking a video lesson as read until full playback is detected.

> **Pro-Tip:** Do not waste time waiting through the video at standard 1x playback. Move directly to Step 21 to apply the scrubber fast-forward shortcut.

Step 21 — Fast-Forward Video Timeline to Unlock and Advance

Drag the scrubber timeline to the very end (-0:00) to unlock completion and advance



[21.1] Video Scrubber Timeline: Drag the video playback scrubber handle all the way to the right end of the timeline (until the timer displays '-0:00'). Fast-forwarding to the end immediately triggers the player's video completion event.

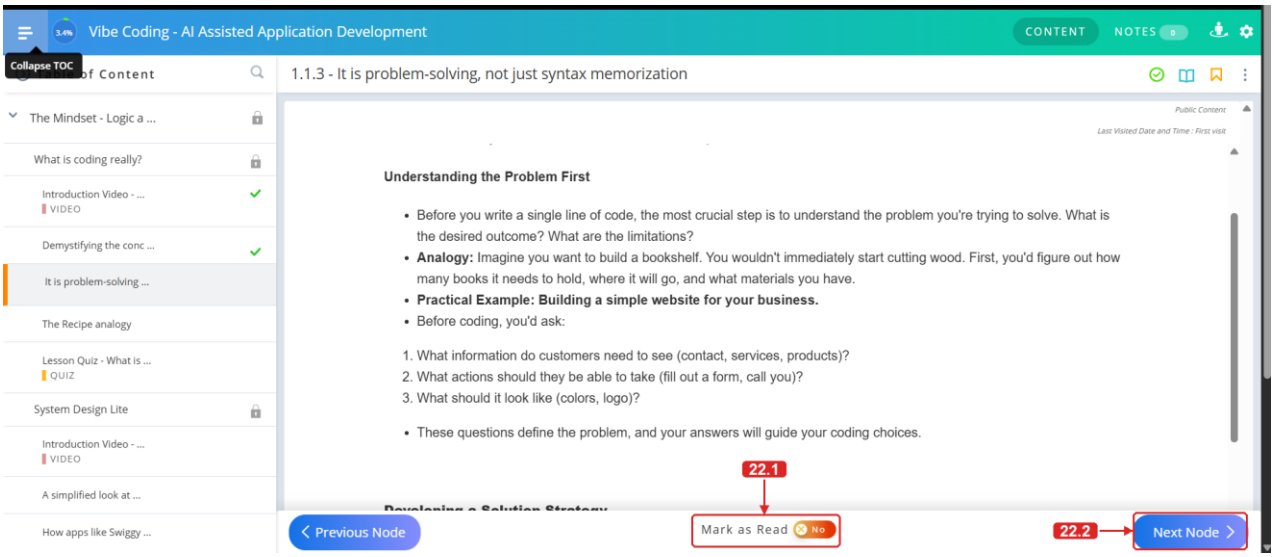
[21.2] Marked as Read: Yes Status: Once the video is at the end, the bottom status unlocks and confirms 'Marked as Read: Yes' in green.

[21.3] Next Node > Button: Click the active blue 'Next Node >' button at the bottom right to advance immediately to the next module node.

> **Pro-Tip:** Video lessons require playback completion to satisfy the platform's video watch trigger. Scrubbing directly to the last second fulfills this requirement instantly, eliminating 15 to 30 minutes of passive playback per video.

Step 22 — Text Reading Nodes: Immediately Toggle 'Mark as Read: Yes'

Pure text reading nodes have no video/timer lock—toggle 'Mark as Read: Yes' immediately and advance



[22.1] Mark as Read Toggle Switch: On text and conceptual reading nodes (such as Lesson 1.1.3 reading notes), there is no video player or minimum time lock. Click the toggle switch directly to change 'Mark as Read: No' to 'Marked as Read: Yes' immediately without waiting.

[22.2] Next Node > Button: Click the blue 'Next Node >' button at the bottom right right away to proceed without spending time reading through the article text.

> **Pro-Tip:** Unlike video nodes which enforce timeline completion, text nodes allow instant manual marking. You can toggle the switch to Yes and click 'Next Node >' within one second of the page opening.

Step 23 — Complete Lesson Quiz & Advance Course Progress

Submit the Lesson Quiz, click 'View Result' and 'Continue' to finalize module progress



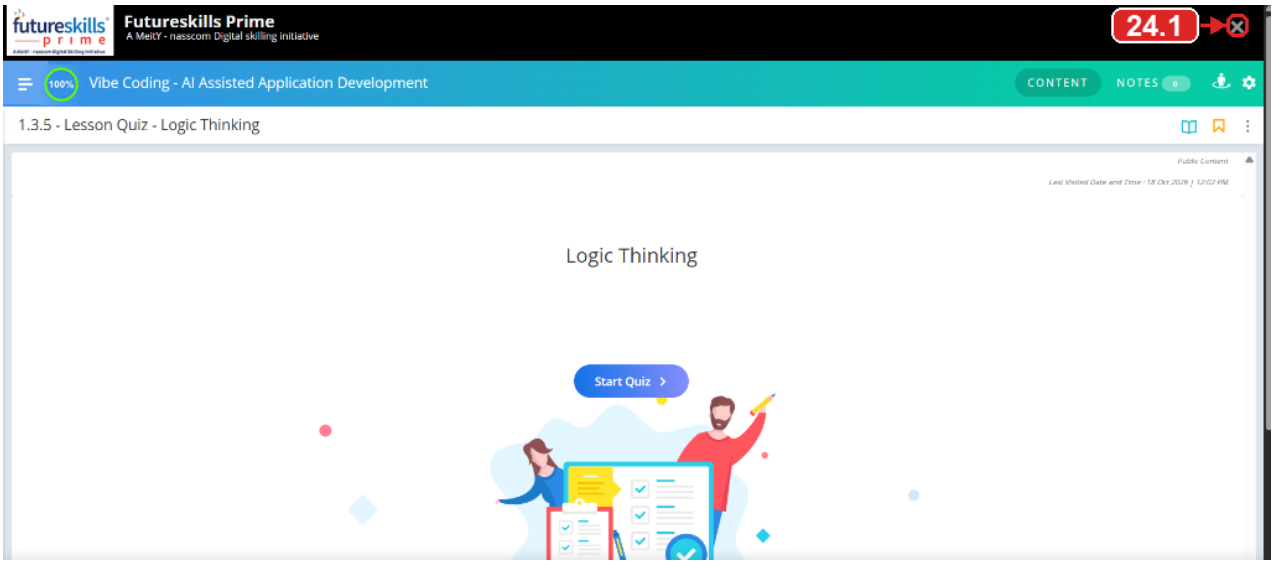
[23.1] Quiz Checkpoint Screen: Lesson 1.1.5 opens the Chapter 1 Quiz ('Lesson Quiz - What is Coding Really').

[23.2] Start Quiz Button: Click the blue 'Start Quiz >' button to begin the assessment.

> **Pro-Tip:** Answer the multiple-choice questions and submit. On the score summary screen, click 'View Result' and then click 'Continue' to register completion and advance your progress to 100% across all required chapters.

Step 24 — Exit Course Player via Header 'X' Icon

Click the 'X' (Exit) icon in the top right corner of the course player header to exit

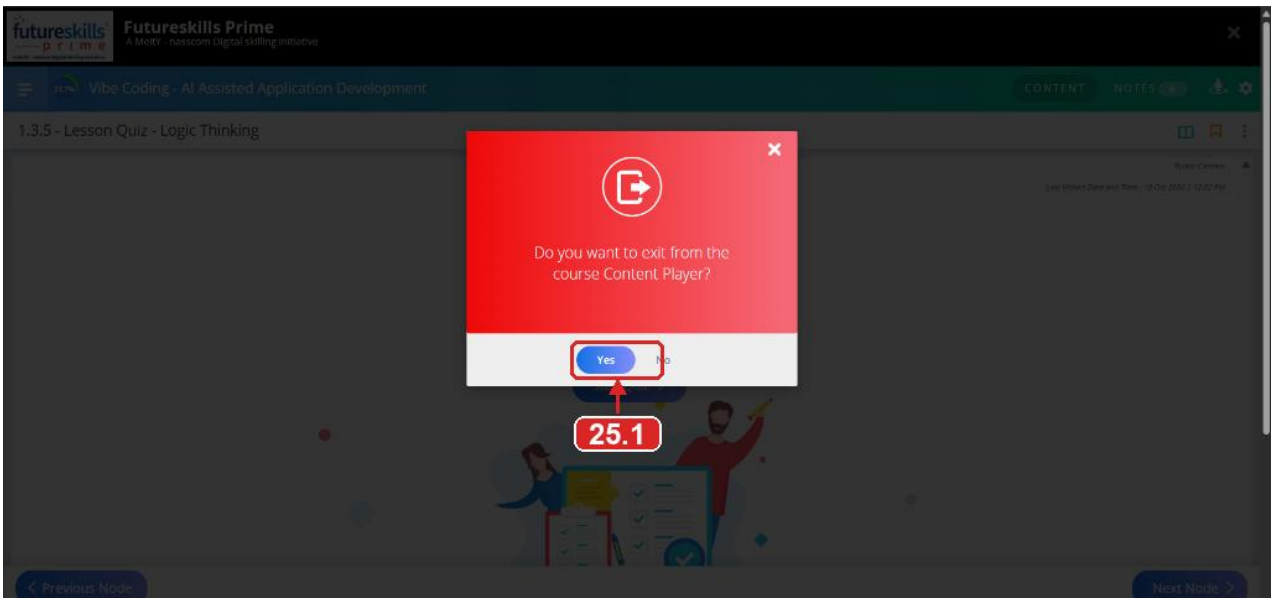


[24.1] Header Exit 'X' Icon: In the top right corner of the black course player header bar, click the 'X' exit icon.

> **Pro-Tip:** Do not close the browser tab directly. Clicking the 'X' icon inside the player safely preserves your module progress and triggers the exit sequence back to your student dashboard.

Step 25 — Confirm Exit in Content Player Dialog

In the exit confirmation dialog, click 'Yes' to return to your student dashboard

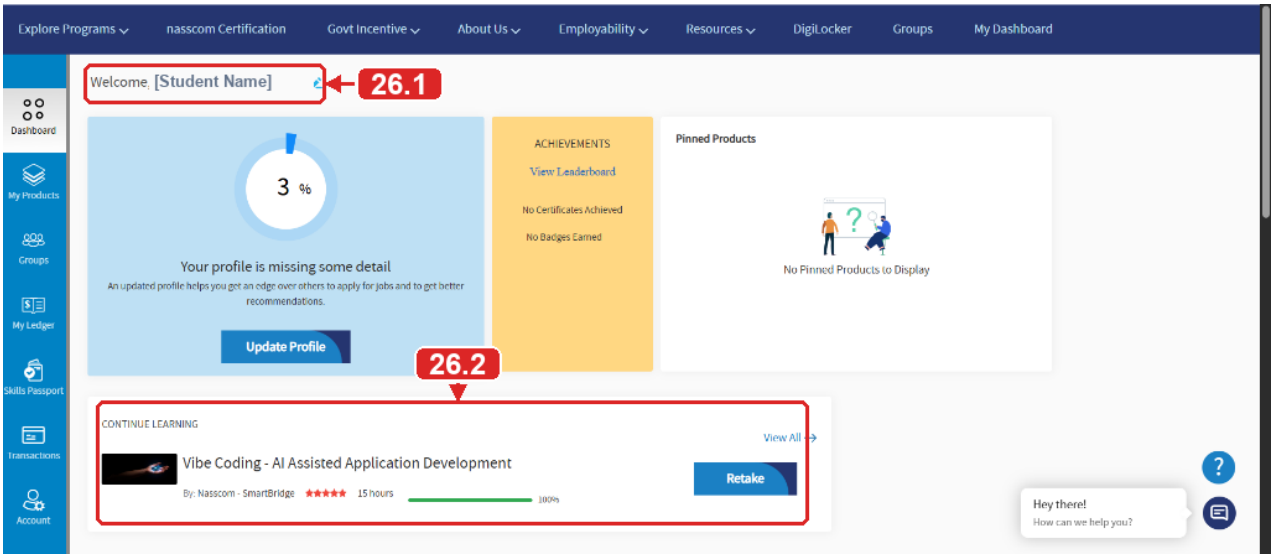


[25.1] 'Yes' Confirmation Button: When prompted with 'Do you want to exit from the course Content Player?', click the blue pill 'Yes' button.

> **Pro-Tip:** Confirming exit redirects you directly to the FutureSkills Prime dashboard where your completed course card and student profile are immediately accessible.

Step 26 — Capture 100% Course Completion Dashboard & Convert to PDF

Verify 100% course completion on your FutureSkills Prime dashboard, capture a full-screen screenshot, and convert it to a PDF document



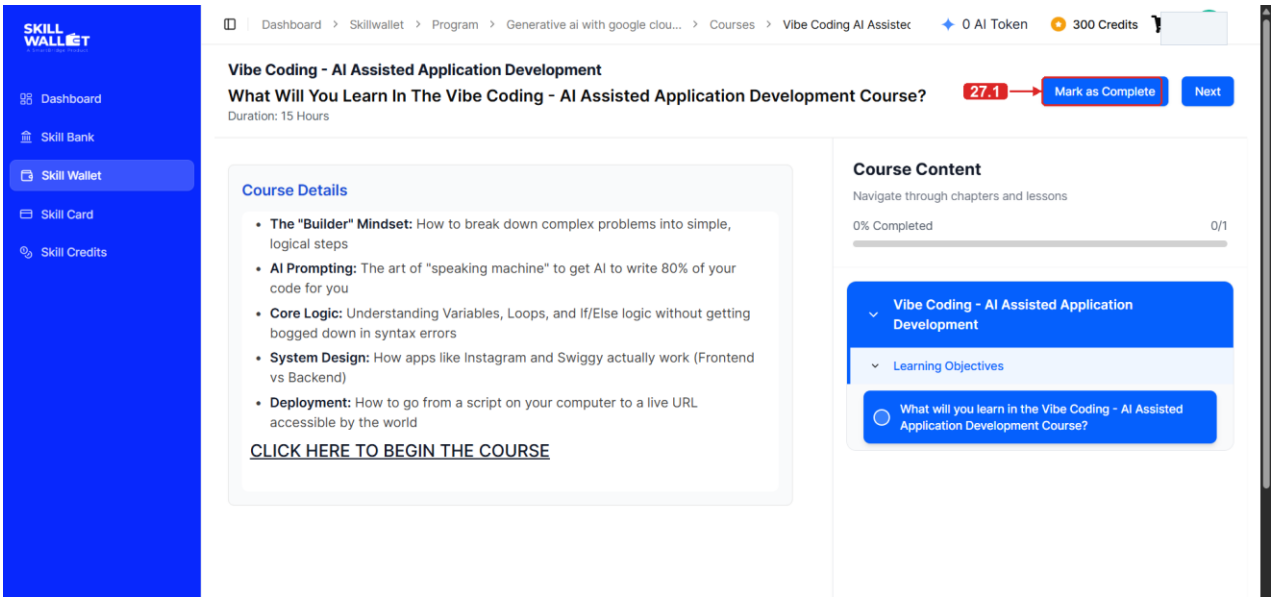
[26.1] Student Profile / Name Verification Area: Confirm your registered student greeting (e.g. 'Welcome, [Your Name]') is clearly displayed in the top header banner.

[26.2] 100% Progress Bar & Retake Button: Under the 'CONTINUE LEARNING' section, locate the 'Vibe Coding - AI Assisted Application Development' card showing the complete green 100% progress bar and the 'Retake' status button.

> **Pro-Tip:** Capture your full-screen screenshot (Win + Shift + S or PrtScn) showing BOTH your student name greeting and the 100% progress card with the 'Retake' button. MANDATORY PDF CONVERSION: Skill Wallet strictly enforces '(PDF only)' and rejects image uploads (.png/.jpg). Convert your screenshot to PDF immediately: Right-click the screenshot image file > select Print > choose 'Microsoft Print to PDF' as the printer > click Print and save as 'Course_Completion_Proof.pdf' (or paste the screenshot into MS Word / Google Docs and choose File > Save As PDF).

Step 27 — Return to Skill Wallet and Click 'Mark as Complete'

Return to Skill Wallet and click 'Mark as Complete' in the top right

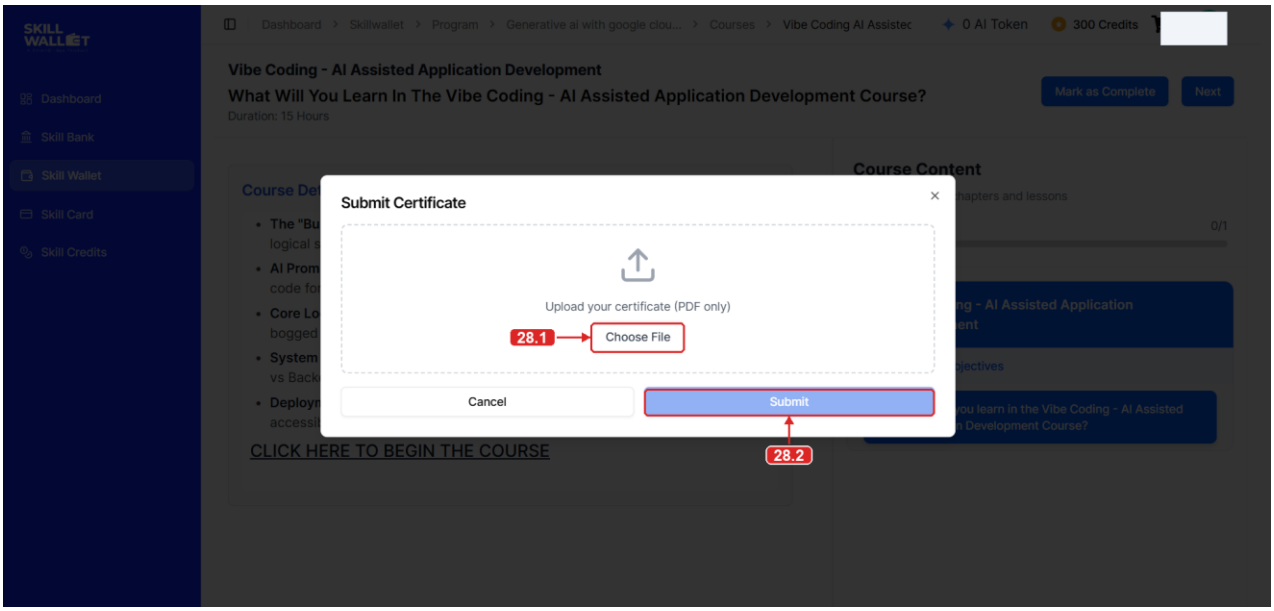


[27.1] Top-Right 'Mark as Complete' Button: Switch back to your Skill Wallet browser tab on the Vibe Coding course page and click the solid blue 'Mark as Complete' button in the upper right corner.

> **Pro-Tip:** Ensure your converted 'Course_Completion_Proof.pdf' is saved and ready on your computer before clicking. Clicking 'Mark as Complete' immediately opens the PDF certificate upload dialog.

Step 28 — Upload Converted Completion PDF and Submit

Upload your converted course completion PDF to Skill Wallet and submit to finalize your credit record



[28.1] 'Choose File' Button: In the 'Upload your certificate (PDF only)' modal, click 'Choose File' and select your converted completion PDF ('Course_Completion_Proof.pdf'). Note: Only .pdf files are accepted by the portal; image files (.png/.jpg) will be rejected.

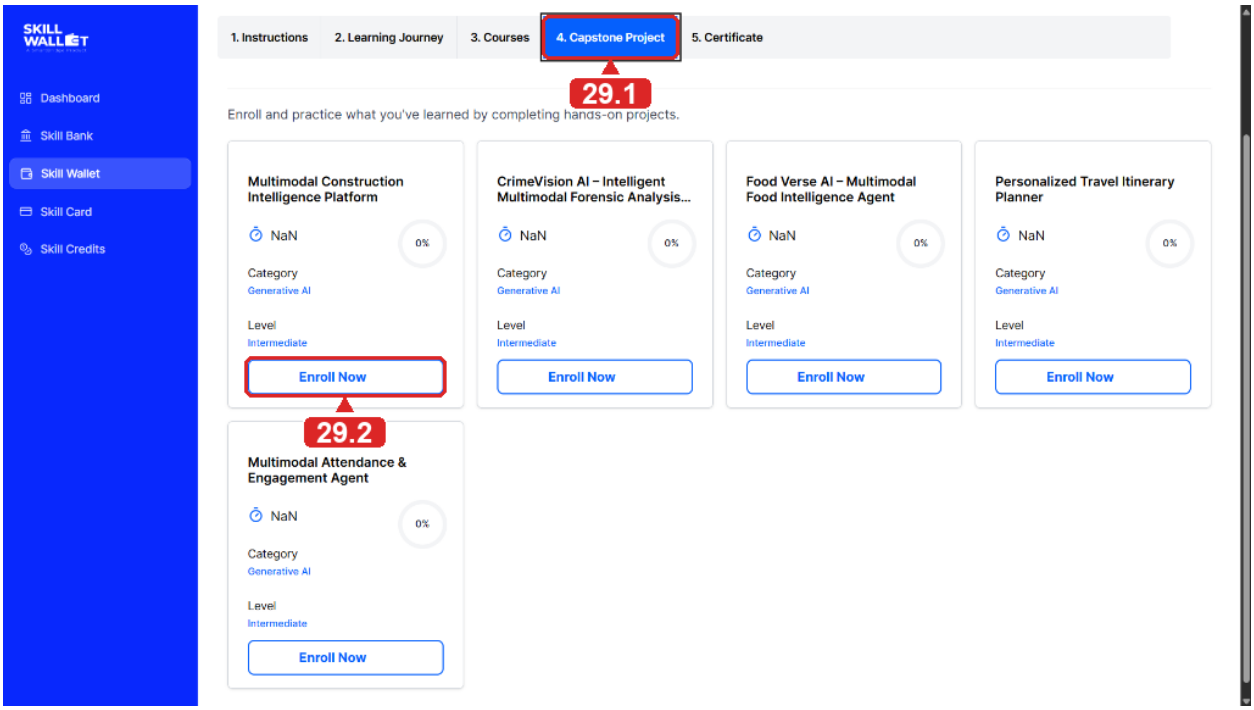
[28.2] 'Submit' Button: Click the solid blue 'Submit' button to complete your verification submission.

> **Pro-Tip:** Mandatory PDF Requirement: The submission modal strictly enforces '(PDF only)'. Upload your converted PDF containing your 100% dashboard completion screenshot. You do NOT need to wait for or request a separate NASSCOM completion certificate—the 100% completion dashboard PDF satisfies all college credit requirements. Once submitted, your course status is marked 100% complete and your mandatory credit record is officially secured.

Step 29 — Navigate to Capstone Projects & Select Starting Module

Access Tab '4. Capstone Project' on Skill Wallet and select any of the 5 required AI Capstone projects to begin

> ⚠ **Note: STUDENT CHOICE: You may choose to start with ANY of the 5 projects. In this guide, Project 1 ('Multimodal Construction Intelligence Platform') is annotated as the recommended fast-track starting module, but all 5 projects must ultimately be completed.**



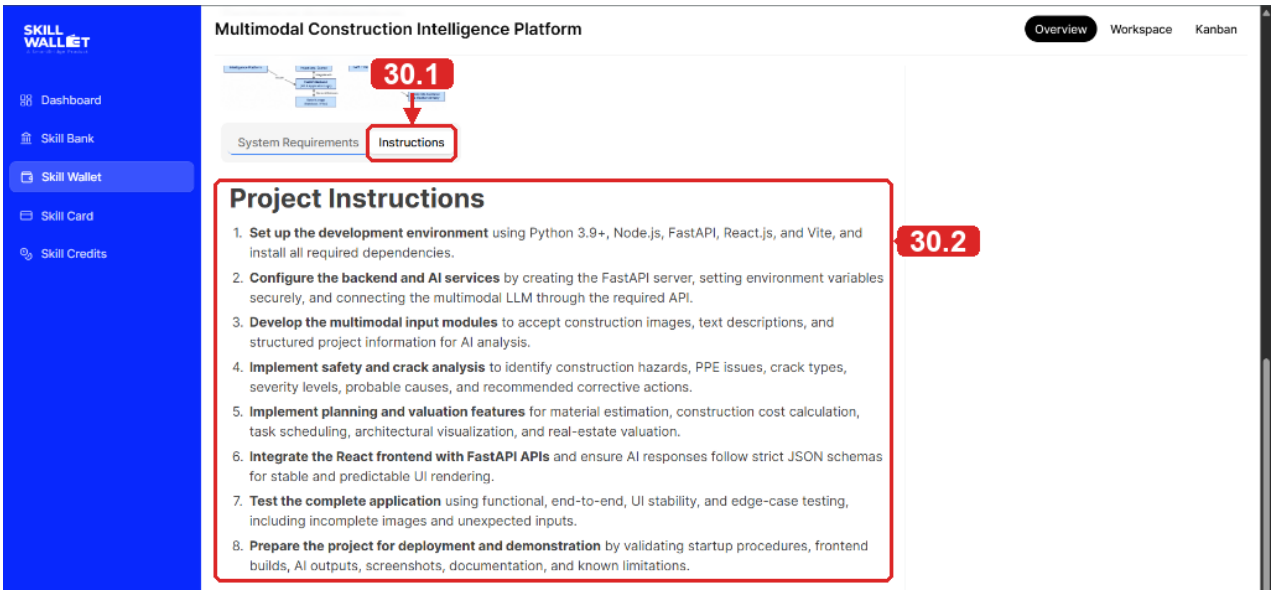
[29.1] '4. Capstone Project' Tab: In the top navigation bar, click the '4. Capstone Project' tab to view your assigned Generative AI capstone project catalog.

[29.2] 'Enroll Now' Button: Click 'Enroll Now' on your selected project card (e.g. Multimodal Construction Intelligence Platform) to activate the project workspace.

> 💡 **Pro-Tip:** All 5 projects require Generative AI multimodal intelligence. You can complete them in any order. Project 1 is highlighted here as the fastest reference template. To unlock your final degree credits, you will repeat the submission process for all 5 projects.

Step 30 — Review Architecture Specifications & 8-Point Project Instructions

Review the official technical requirements, data contracts, and implementation milestones before building your AI solution



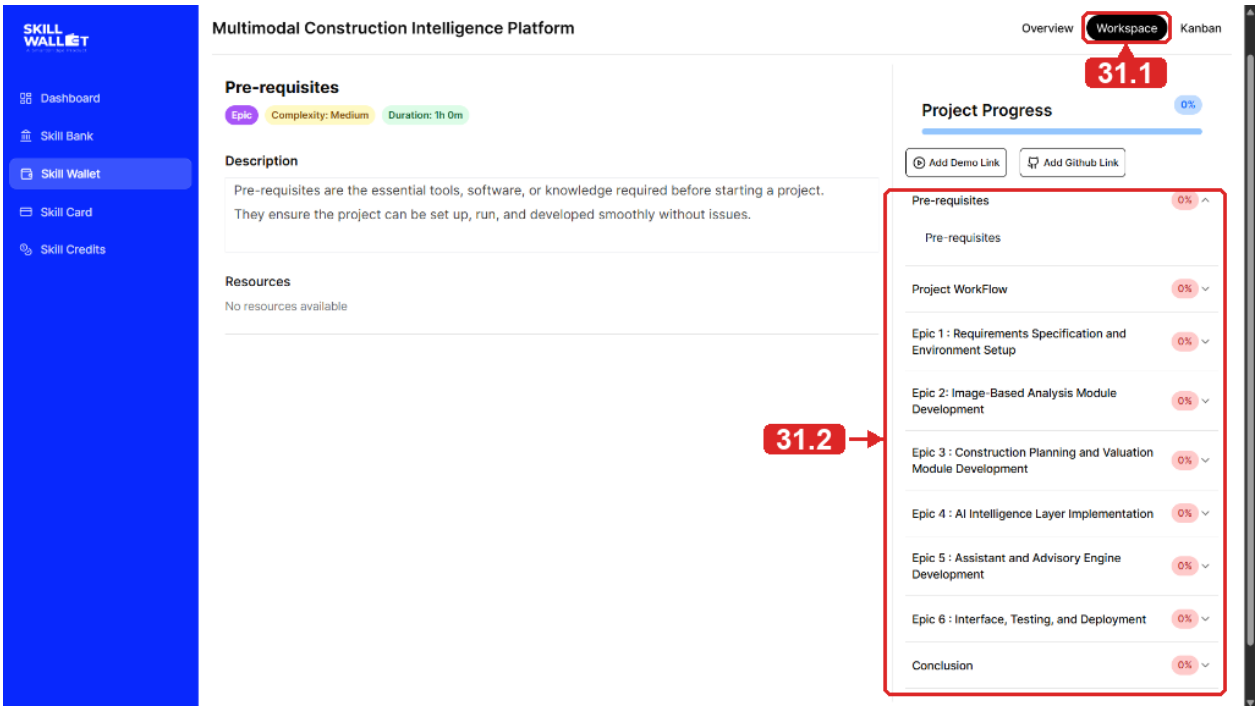
[30.1] 'Instructions' Tab Button: Click the 'Instructions' tab next to 'System Requirements' to reveal the 8-point implementation specification.

[30.2] 8-Point Project Instructions: Note the core requirements: environment setup, backend AI services, multimodal input modules, domain intelligence analysis, JSON data contracts, and deployment validation.

> **Pro-Tip:** Do not attempt to write complex boilerplate code manually from scratch. Use modern 'Vibe Coding' AI assistants (Cursor, Claude, ChatGPT, Gemini) with the master prompts provided in Section 7.2 of this guide. These prompts auto-generate fully functioning Python + Streamlit applications conforming strictly to all 8 instructions.

Step 31 — Access Project Workspace & Track Epic Milestone Progress

Open the 'Workspace' tab to review granular requirements across all 6 Epics and track milestone percentages



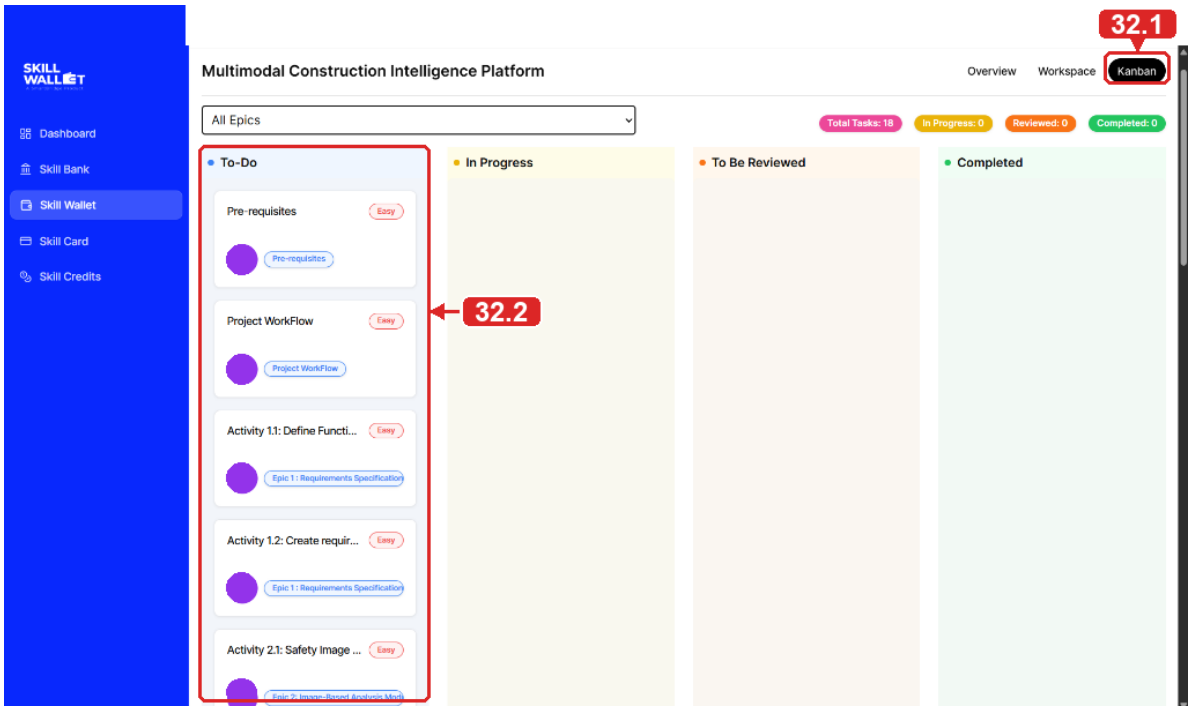
[31.1] 'Workspace' Tab Button: In the upper right navigation of the project header, click the black pill 'Workspace' tab.

[31.2] Project Workflow & Epic Progress Accordion: Expand the right-hand accordion to inspect requirements across Pre-requisites, Project Workflow, Epics 1 through 6, and Conclusion.

> **Pro-Tip:** The Workspace accordion provides a structural roadmap of your project. Each of the 6 Epics maps directly to modules in your AI-generated codebase (e.g. Epic 2 = Safety & Crack Analysis; Epic 5 = Assistant & Advisory Engine).

Step 32 — Manage Milestone Tasks on the Kanban Board

Open the 'Kanban' tab to track the 18 specific milestone activities across To-Do, In Progress, and Completed stages



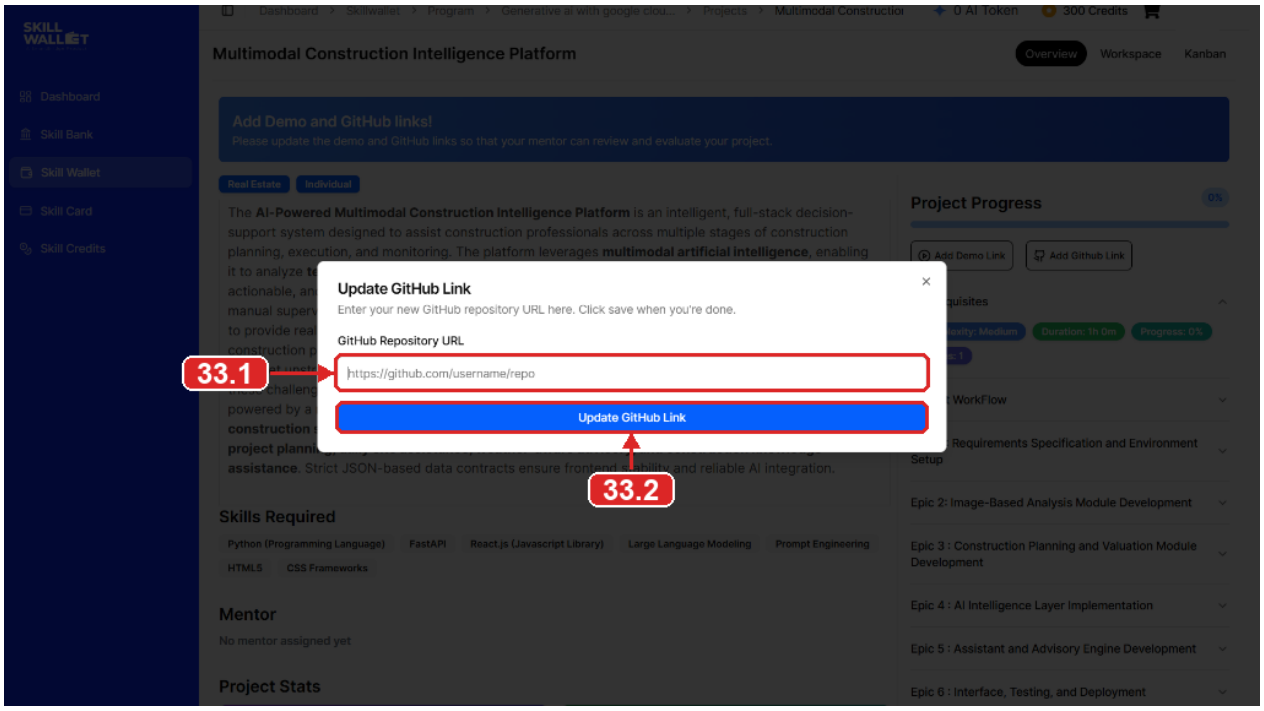
[32.1] 'Kanban' Tab Button: In the top right header, click the black pill 'Kanban' tab to view your interactive Agile task board.

[32.2] To-Do Column & Activity Cards: Review the 18 milestone cards organized by Epic (e.g. Activity 1.1 Requirements, Activity 2.1 Safety Analysis, Activity 3.1 Planning Logic). Move cards to 'Completed' as you assemble your code.

> **Pro-Tip:** Updating tasks on the Kanban board logs your active development progress for your mentor. Completing tasks across Epics 1 to 6 drives your overall Project Progress bar from 0% toward 100%.

Step 33 — Submit GitHub Repository Link via Modal Dialog

Submit your project's GitHub repository URL to Skill Wallet for mentor evaluation



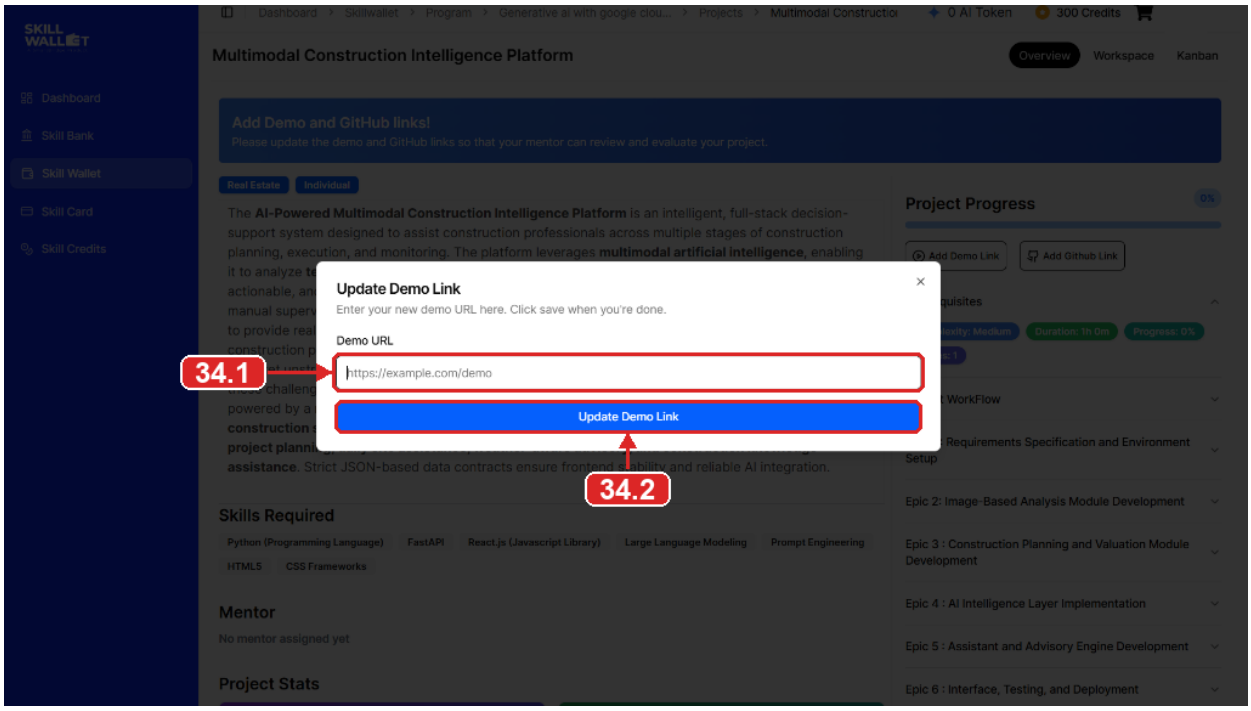
[33.1] 'GitHub Repository URL' Input Field: Click 'Add Github Link' on the project dashboard and paste your public GitHub repository URL (e.g. https://github.com/your-username/your-project-repo).

[33.2] 'Update GitHub Link' Button: Click the solid blue 'Update GitHub Link' button to register your repository with your mentor.

> **Pro-Tip:** Ensure your GitHub repository is public and contains your app.py, requirements.txt, and a clean README.md explaining the features, tech stack, and how to run the project. Mentors verify your repository structure.

Step 34 — Submit Live Hosted Demo URL via Modal Dialog

Deploy your AI application to cloud hosting and submit the live HTTPS demo link for evaluator testing



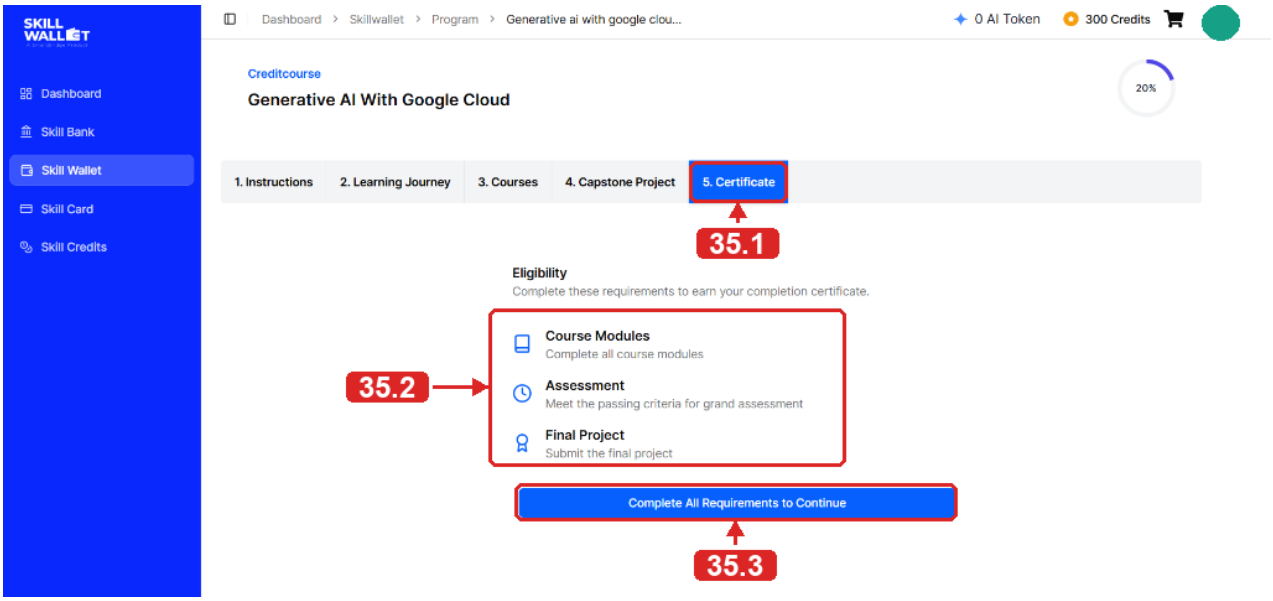
[34.1] 'Demo URL' Input Field: Click 'Add Demo Link' on the project dashboard and paste your live deployed web application URL (e.g. https://your-project.streamlit.app).

[34.2] 'Update Demo Link' Button: Click the solid blue 'Update Demo Link' button to finalize your live demonstration submission.

> **Pro-Tip:** You can host your Python Streamlit application for 100% FREE in under 60 seconds using Streamlit Community Cloud (share.streamlit.io). Simply connect your GitHub account, select your repo and app.py, and click Deploy. Copy the live HTTPS link and submit it here.

Step 35 — Navigate to Tab 5 (Certificate) & Claim Course Credential

Verify your 3-point eligibility checklist to unlock and download your official accredited course completion certificate



[35.1] Tab '5. Certificate' Navigation: After completing all course modules, passing the grand assessment, and submitting your capstone project links, click Tab '5. Certificate' in the top navigation bar.

[35.2] 3-Point Eligibility Checklist: Verify that all 3 mandatory criteria reflect completion: (1) Course Modules completed, (2) Assessment passing criteria met, and (3) Capstone Final Project submitted.

[35.3] 'Complete All Requirements to Continue' Button: Click the solid blue button once all 3 pillars are satisfied to generate and download your official NASSCOM FutureSkills Prime & Skill Wallet accredited completion certificate.

> **Pro-Tip:** Ensure your project's GitHub repository and live Streamlit demo URL are submitted in Steps 33 and 34 before accessing Tab 5. Once the 3 eligibility items are verified, the button activates, allowing you to instantly download your tamper-proof PDF certificate and register the verified credential to your digital Skill Wallet.

APPENDIX: THE 'VIBE CODING' AI CAPSTONE PLAYBOOK

Official Specifications, Architecture Rubrics, and Turnkey Master Prompts for All 5 Capstone Projects

Capstone Strategy & Ground-Truth Curriculum Alignment:
Students are required to complete and submit exactly ONE capstone project on the Skill Wallet portal to achieve 100% course certification. The curriculum specifications below represent the verified, ground-truth platform requirements extracted directly from the authenticated Skill Wallet database. Each project features an architectural breakdown across all Epics and a production-grade master prompt. Students can copy and paste the master prompt directly into an AI coding assistant (Cursor IDE, Claude, ChatGPT, or Antigravity) to generate a fully functioning application that satisfies 100% of the portal's evaluation rubric.

Project 1: Multimodal Construction Intelligence Platform

Category: Generative AI | Real Estate Decision Support | Platform ID: 6ac4c98d1dcde6d7857c1642

Objective & Scope: An intelligent decision-support system analyzing site photos to audit PPE safety compliance, classify concrete cracks with severity scores, estimate construction costs and real estate valuation in INR, and provide weather-aware daily protocols.

Kanban Epics Coverage: Pre-requisites (Python 3.9+, FastAPI, React) | Epic 1 (Requirements & Env) | Epic 2 (PPE Safety Vision & Crack Diagnostics) | Epic 3 (Project Planning & Real Estate Valuation) | Epic 4 (Multimodal AI & JSON Contracts) | Epic 5 (Site Chatbot & Weather Advisory) | Epic 6 (Interface, Testing, Cloud Deployment)

Recommended Tech Stack: Python 3.9+, Streamlit / FastAPI + React, Multimodal LLM (Gemini 1.5 / LLaMA Vision), Pillow, Pollinations/Flux API, JSON Data Contracts

TURNKEY MASTER PROMPT (PROJECT 1)

```
Act as an expert full-stack AI engineer. Build a complete, production-ready Streamlit application in Python named "Multimodal Construction Intelligence Platform" (Skill Wallet Capstone Project 1).

Core Requirements & Modules:
1. Sidebar: Secure input for Google Gemini API key (with graceful fallback to demonstration mode if no key is supplied), architecture overview, and feature navigation.
2. Tab 1 - Site Safety & PPE Audit (Epic 2):
   - File uploader for construction site photos (JPG/PNG).
   - Multimodal prompt analyzing worker PPE compliance (helmets, high-vis vests, steel-toe boots, safety harnesses) under OSHA and IS 456 standards.
   - Output structured JSON containing safety_score (0-100), overall_status, ppe_detection checklist, and list of identified site hazards with severity (Low, Medium, High, Critical) and corrective remedies.
   - Render metric cards, status pills, and downloadable JSON report.
3. Tab 2 - Crack Diagnostics & Structural Integrity (Epic 2):
   - File uploader for concrete/wall surfaces.
   - Multimodal prompt classifying defect type (Hairline, Settlement, Structural Shear, Thermal Expansion, Spalling) with estimated crack width (mm), root causes, recommended engineering repairs (epoxy injection, structural re-grouting), and urgency.
4. Tab 3 - Planning & Real Estate Valuation (Epic 3):
   - Interactive inputs for structure type, built-up sqft, location tier (Tier 1/2/3 Indian cities), and completion percentage slider.
   - Calculate total project budget, capital expenditure incurred, current market asset valuation in INR (Lakhs/Crores), and steel/cement requirement metrics with milestone progress bar.
   - Conceptual 3D architectural rendering integration via Pollinations AI prompt generator.
5. Tab 4 - Daily Site Assistant & Weather-Aware Advisory (Epic 5):
   - Weather condition selector (High Wind, Heavy Rain, Extreme Heat, Normal) displaying dynamic safety alerts (e.g. suspending crane operations, electrical hazard mitigation, concrete curing protocol).
   - Interactive multi-turn construction engineering assistant chatbot.

Ensure clean, modular code with exception handling, modern CSS styling, and zero external database dependencies for seamless 1-click deployment on Streamlit Community Cloud.
```

Project 2: CrimeVision AI – Intelligent Multimodal Forensic Analysis Platform

Category: Generative AI | Criminal Justice & Public Safety | Platform ID: 6ac4c9851dcde6d7857c163b

Objective & Scope: An automated forensic intelligence platform fusing image, video, and audio evidence to detect people/weapons, transcribe witness audio via Whisper, reconstruct chronological crime timelines, provide grounded AI evidence Q&A, and generate court-admissible PDF reports.

Kanban Epics Coverage: Pre-requisites (Flask/PyTorch/Whisper) | Epic 1 (Initialization & Env) | Epic 2 (RT-DETR Object Detection & Scene Captioning) | Epic 3 (Whisper Audio Transcription & Timeline Extraction) | Epic 4 (Video Frame Tracking & Crime Classification) | Epic 5 (Multimodal Timeline Fusion & Evidence Chat) | Epic 6 (Dashboard, Testing, ReportLab PDF Export)

Recommended Tech Stack: Python 3.9+, Streamlit / Flask, RT-DETR / YOLOv8, OpenAI Whisper, Groq / Gemini LLM, ReportLab, OpenCV, JSON

TURNKEY MASTER PROMPT (PROJECT 2)

```
Act as a senior forensic systems architect. Build a complete, production-ready Streamlit application in Python named "CrimeVision AI - Intelligent Multimodal Forensic Analysis Platform" (Skill Wallet Capstone Project 2).

Core Requirements & Modules:
1. Configuration & Security:
   - Secure API key input (Groq or Gemini) with automatic demo fallback mode.
   - Forensic chain-of-custody tracking indicators (Evidence ID, Timestamp, Investigator Sign-off).
2. Tab 1 - Crime Scene Visual Evidence Tagger (Epic 2):
   - Image uploader for crime scene photography.
   - Multimodal object detection simulating RT-DETR identifying persons, firearms, edged weapons, blood spatter, and forced entry points.
   - Output structured forensic analysis: Threat Level (Low, Medium, High, Critical), Scene Classification (Burglary, Homicide, Vandalism, Assault), detected items with confidence scores and spatial coordinates, and prioritized investigative leads.
3. Tab 2 - Witness Audio Transcription & Timeline (Epic 3):
   - Audio uploader for witness statements or emergency 911/112 calls.
   - Audio transcription analysis (Whisper API / fallback) extracting key events, speaker tags, and timestamped chronological markers.
4. Tab 3 - CCTV Video Analysis & Event Progression (Epic 4):
   - Video file processor sampling frames and tracking suspect trajectories and vehicle movements across seconds.
   - Chronological event log table with classified legal penal crime categories.
5. Tab 4 - Multimodal Fusion & Grounded Evidence Chat (Epic 5):
   - Unified master forensic event ledger combining image, audio, and video timestamps.
   - Grounded RAG conversational interface allowing investigators to interrogate evidence in natural language (e.g., "Was a weapon visible before the suspect fled?") with zero hallucination and strict citation of evidence IDs.
6. Tab 5 - Automated Forensic PDF Export (Epic 6):
   - 1-click automated generation of court-admissible forensic summary PDF reports using ReportLab.

Ensure clean, modular code with exception handling, modern dark-themed forensic styling, and zero external database dependencies for seamless deployment on Streamlit Community Cloud.
```

Project 3: Food Verse AI – Multimodal Food Intelligence Agent

Category: Generative AI | Nutrition, Freshness & Kitchen Safety | Platform ID: 6ac4ca021dcde6d7857c169a

Objective & Scope: A multimodal food intelligence ecosystem analyzing food images, kitchen videos, and dietary profiles to recognize dishes, estimate macro/micronutrient scores, detect food freshness and spoilage, monitor commercial kitchen hygiene, and generate personalized meal plans.

Kanban Epics Coverage: Pre-requisites (FastAPI/React/OpenCV) | Epic 1 (Requirements & Env) | Epic 2 (Food Recognition & Nutrition Breakdown) | Epic 3 (Freshness & Spoilage Detection) | Epic 4 (Kitchen Hygiene Compliance & Smart Recipe Coach) | Epic 5 (TDEE Meal Planner) | Epic 6 (Interface, Edge Cases, Deployment)

Recommended Tech Stack: Python 3.9+, Streamlit / FastAPI + React, OpenCV, PyTorch, Meta LLaMA / Gemini 1.5, Pillow, JSON Data Contracts

TURNKEY MASTER PROMPT (PROJECT 3)

```
Act as an expert food science and AI systems engineer. Build a complete, production-ready Streamlit application in Python named "Food Verse AI - Multimodal Food Intelligence Agent" (Skill Wallet Capstone Project 3).

Core Requirements & Modules:
1. Configuration & Persona:
   - Secure API key input (Gemini or Groq) with standalone demonstration fallback mode.
   - Interactive navigation across FoodVerse core intelligence pillars.
2. Tab 1 - Food Recognition & Nutrition Audit (Epic 2):
   - Image uploader for meal and dish photos.
   - Multimodal vision prompt identifying dish name, cuisine, ingredients, and portion size.
   - Calculate detailed nutritional breakdown: Calories (kcal), Protein (g), Carbohydrates (g), Fats (g), Fiber (g), allergen alerts, and a composite Health Score (1-100).
   - Render interactive macro charts and health badges.
3. Tab 2 - Food Freshness & Spoilage Diagnostic (Epic 3):
   - Image uploader for perishable raw items (fruits, vegetables, meats, dairy).
   - Vision analysis evaluating surface texture, discoloration, wrinkling, and mold.
   - Classify freshness stage: [Fresh | Moderately Fresh | Spoiled].
   - Provide actionable food safety advice (Safe to consume, Consume within 24h, Discard immediately) and proper preservation storage tips.
4. Tab 3 - Kitchen Hygiene & Food Safety Compliance (Epic 4):
   - Image/video uploader for commercial kitchen environments.
   - Detect hygiene indicators: Chef hat/hairnet compliance, glove usage, sanitization state, and cross-contamination hazards (raw poultry near vegetables).
   - Calculate an overall Kitchen Hygiene Compliance Score (%) with itemized violation logs.
5. Tab 4 - Smart Kitchen Recipe Assistant (Epic 4):
   - Pantry inventory input allowing users to list available ingredients.
   - Generate creative, nutritious recipes with cooking time, difficulty level, step-by-step instructions, and ingredient substitutions.
6. Tab 5 - Personalized Meal Planner (Epic 5):
   - User profile form capturing age, gender, weight, activity level, dietary constraints (Vegan, Keto, Diabetic-Friendly), and fitness goals (Fat Loss, Muscle Gain).
   - Automatically compute daily BMR/TDEE caloric targets and generate an adaptive 7-day meal schedule with automated grocery shopping list.

Ensure clean, modular code with exception handling, modern vibrant styling, and zero external database dependencies for seamless 1-click deployment on Streamlit Community Cloud.
```

Project 4: Personalized Travel Itinerary Planner

Category: Generative AI | Intelligent SaaS Trip Planning | Platform ID: 6ac33976fb8fc7965cd9236d

Objective & Scope: A full-stack, AI-powered travel application generating optimized day-by-day vacation itineraries, clustering attractions to eliminate travel transit fatigue, offering vibe-based destination recommendations, and providing exportable travel guides.

Kanban Epics Coverage: Pre-requisites (Node/FastAPI/Supabase) | Epic 1 (Vibe Coding Scaffolding) | Epic 2 (Supabase PostgreSQL Relational Schema) | Epic 3 (Preference Capture & Destination Recommendation) | Epic 4 (Day-Wise Itinerary & Route Optimization) | Epic 5 (Trip Persistence & PDF Export) | Epic 6-7 (Cloud Deployments on Render & Vercel, Final Testing)

Recommended Tech Stack: Python 3.9+, Streamlit / FastAPI + React, Supabase / SQLite Local Fallback, Gemini / Groq LLM, ReportLab, JSON

TURNKEY MASTER PROMPT (PROJECT 4)

```
Act as a senior full-stack travel SaaS architect. Build a complete, production-ready Streamlit application
in Python named "Personalized Travel Itinerary Planner" (Skill Wallet Capstone Project 4).

Core Requirements & Modules:
1. Configuration & User Preferences (Epic 3):
   - Secure API key input (Gemini or Groq) with demo mode.
   - Comprehensive trip preference form: Destination (or open-ended vibe: Beach, Mountains, Cultural,
Urban), Trip Duration (days), Travel Dates, Budget Level (Budget, Moderate, Luxury), Travel Party (Solo,
Couple, Family, Friends), Pace (Relaxed, Moderate, Fast-Paced), and Interests (Food, Adventure, History,
Nightlife).
2. Tab 1 - Smart Destination Recommender (Epic 3):
   - For users without a fixed destination, recommend top 3 matching cities based on budget, weather season,
and travel mood with highlight tags.
3. Tab 2 - Day-by-Day Optimized Itinerary Generator (Epic 4):
   - Generate structured day-wise plans with sequential time slots (Morning Sightseeing, Lunch
Recommendation, Afternoon Leisure, Evening Dining & Entertainment).
   - Geographic clustering logic: Automatically grouping nearby attractions on the same day to minimize
transit fatigue and prevent backtracking across cities.
   - Itemized estimated costs per activity and running cumulative budget tracker in local currencies and
USD/INR.
4. Tab 3 - Vibe Customization & Dynamic Activity Swap (Epic 4):
   - Interactive controls to modify specific slots (e.g. "Swap museum for outdoor market", "Find vegetarian
lunch spot") and regenerate single days without wiping the rest of the plan.
5. Tab 4 - Trip Management & Offline PDF Export (Epic 5):
   - Simulated database persistence saving multiple trips with trip history viewer.
   - 1-click export generating a clean, beautifully formatted offline PDF travel itinerary guide with
emergency contact checklists.

Ensure clean, modular code with exception handling, modern wanderlust-inspired CSS styling, and zero
external database dependencies for seamless 1-click deployment on Streamlit Community Cloud.
```

Project 5: Multimodal Attendance & Engagement Agent

Category: Generative AI | Privacy-First Classroom Analytics | Platform ID: 6ac775fb32e2e6be34c17468

Objective & Scope: An offline, privacy-preserving classroom intelligence system fusing computer vision (MediaPipe face tracking for sustained presence attendance) and audio signal processing (speech activity vs ambient distraction) to compute engagement scores and generate real-time teaching advisories.

Kanban Epics Coverage: Pre-requisites (OpenCV/MediaPipe/Librosa) | EPIC 1 (Requirements & Offline Env) | EPIC 2 (Face Tracking & Sustained Presence Attendance) | EPIC 3 (Audio Speech & Noise Processing) | EPIC 4 (Multimodal Signal Fusion & Engagement Scoring) | EPIC 5 (Teaching Suggestion Engine & Session Reports) | EPIC 6 (Streamlit Telemetry Dashboard, Testing)

Recommended Tech Stack: Python 3.9+, Streamlit, OpenCV, MediaPipe, Librosa / Acoustic Signal Processing, Pandas, Matplotlib, JSON

TURNKEY MASTER PROMPT (PROJECT 5)

```
Act as an expert AI systems and computer vision engineer. Build a complete, production-ready Streamlit
application in Python named "Multimodal Attendance & Engagement Agent" (Skill Wallet Capstone Project 5).

Core Requirements & Modules:
1. Privacy-First System Principles:
   - 100% offline edge processing: Zero external cloud video transmission to guarantee classroom privacy.
   - Ethical telemetry design: Focus on collective pedagogical feedback rather than punitive individual
surveillance.
2. Tab 1 - Automated Presence-Based Attendance (Epic 2):
   - Video feed processor (supporting webcam or uploaded classroom video/photo).
   - Participant face detection and tracking ID assignment (OpenCV / MediaPipe).
   - Sustained Presence Logic: Require participants to be detected across a sustained duration threshold
(>70% session time) to prevent false attendance marks from temporary hallway pass-throughs.
   - Exportable attendance register table (CSV download) with Student ID, First Seen, Last Seen, Total
Duration, and Status (Present/Absent).
3. Tab 2 - Acoustic Interaction & Noise Dynamics (Epic 3):
   - Audio stream processor segmenting signals into time windows.
   - Spectral energy and Voice Activity Detection (VAD) classifying audio into: Silence, Instructor Lecture,
Student Discussion / Verbal Interaction, and High Ambient Classroom Distraction.
4. Tab 3 - Multimodal Fusion & Engagement Scoring (Epic 4):
   - Fuse visual attentiveness indicators (forward head pose, upright posture vs distraction/slumping) and
audio interaction energy into a unified time-step matrix.
   - Compute normalized Engagement Index (0-100) and plot real-time timeline engagement curves showing
attention peaks and dips.
5. Tab 4 - AI Teaching Suggestion Engine (Epic 5):
   - Heuristic advisory agent generating real-time, non-intrusive pedagogical prompts when attention flags
(e.g. "Collective engagement dropped below 45% for 5 minutes — recommend launching a quick poll, interactive
discussion, or a 2-minute stretch break").
6. Tab 5 - Session Analytics & Educator Report (Epic 5 & 6):
   - Comprehensive post-session summary dashboard rendering key session KPIs, participation rates, and
downloadable session summary report.

Ensure clean, modular code with exception handling, modern clean educational dashboard styling, and zero
external database dependencies for seamless 1-click deployment on Streamlit Community Cloud.
```

1-Click Cloud Deployment & Final Proof Registration

To successfully satisfy Step 33 (GitHub Repository Submission) and Step 34 (Live Hosted Demo URL Registration) on Skill Wallet:

1. Create a GitHub Repository: Push your generated app.py, requirements.txt, and README.md to a new public GitHub repo.
2. Deploy for Free to Streamlit Community Cloud: Go to share.streamlit.io, click 'New app', select your repo, branch (main), and main file path (app.py), then click Deploy.
3. Register Deliverables on Skill Wallet: In Step 33, paste your GitHub repository link. In Step 34, paste your live public URL (e.g. https://<app-name>.streamlit.app). This unlocks the 100% course completion certificate.